```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.90                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    AUG  1, 2026                *
*           Time=    23:31:49                    *
*                                                *
*      USER ID: FSOS-X R9 EXP                    *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: E:\Staad pro v8i projects\G+4\G+4.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 01-AUG-26
    4. END JOB INFORMATION
    5. INPUT WIDTH 79
    6. UNIT METER KN
    7. JOINT COORDINATES
    8. 1 0 0 0; 2 3 0 0; 3 6 0 0; 4 9 0 0; 5 12 0 0; 6 15 0 0; 7 0 3.3 0
    9. 8 3 3.3 0; 9 6 3.3 0; 10 9 3.3 0; 11 12 3.3 0; 12 15 3.3 0; 13 0 6.3 0
   10. 14 3 6.3 0; 15 6 6.3 0; 16 9 6.3 0; 17 12 6.3 0; 18 15 6.3 0
   11. 19 0 9.3 0; 20 3 9.3 0; 21 6 9.3 0; 22 9 9.3 0; 23 12 9.3 0
   12. 24 15 9.3 0; 25 0 12.3 0; 26 3 12.3 0; 27 6 12.3 0; 28 9 12.3 0
   13. 29 12 12.3 0; 30 15 12.3 0; 31 0 15.3 0; 32 3 15.3 0; 33 6 15.3 0
   14. 34 9 15.3 0; 35 12 15.3 0; 36 15 15.3 0; 37 0 0 3; 38 3 0 3; 39 6 0 3
   15. 40 9 0 3; 41 12 0 3; 42 15 0 3; 43 0 3.3 3; 44 3 3.3 3; 45 6 3.3 3
   16. 46 9 3.3 3; 47 12 3.3 3; 48 15 3.3 3; 49 0 6.3 3; 50 3 6.3 3
   17. 51 6 6.3 3; 52 9 6.3 3; 53 12 6.3 3; 54 15 6.3 3; 55 0 9.3 3
   18. 56 3 9.3 3; 57 6 9.3 3; 58 9 9.3 3; 59 12 9.3 3; 60 15 9.3 3
   19. 61 0 12.3 3; 62 3 12.3 3; 63 6 12.3 3; 64 9 12.3 3; 65 12 12.3 3
   20. 66 15 12.3 3; 67 0 15.3 3; 68 3 15.3 3; 69 6 15.3 3; 70 9 15.3 3
   21. 71 12 15.3 3; 72 15 15.3 3; 73 0 0 6; 74 3 0 6; 75 6 0 6; 76 9 0 6
   22. 77 12 0 6; 78 15 0 6; 79 0 3.3 6; 80 3 3.3 6; 81 6 3.3 6; 82 9 3.3 6
   23. 83 12 3.3 6; 84 15 3.3 6; 85 0 6.3 6; 86 3 6.3 6; 87 6 6.3 6
   24. 88 9 6.3 6; 89 12 6.3 6; 90 15 6.3 6; 91 0 9.3 6; 92 3 9.3 6
   25. 93 6 9.3 6; 94 9 9.3 6; 95 12 9.3 6; 96 15 9.3 6; 97 0 12.3 6
   26. 98 3 12.3 6; 99 6 12.3 6; 100 9 12.3 6; 101 12 12.3 6; 102 15 12.3 6
   27. 103 0 15.3 6; 104 3 15.3 6; 105 6 15.3 6; 106 9 15.3 6; 107 12 15.3 6
   28. 108 15 15.3 6; 109 0 0 9; 110 3 0 9; 111 6 0 9; 112 9 0 9; 113 12 0 9
   29. 114 15 0 9; 115 0 3.3 9; 116 3 3.3 9; 117 6 3.3 9; 118 9 3.3 9
   30. 119 12 3.3 9; 120 15 3.3 9; 121 0 6.3 9; 122 3 6.3 9; 123 6 6.3 9
   31. 124 9 6.3 9; 125 12 6.3 9; 126 15 6.3 9; 127 0 9.3 9; 128 3 9.3 9
   32. 129 6 9.3 9; 130 9 9.3 9; 131 12 9.3 9; 132 15 9.3 9; 133 0 12.3 9
   33. 134 3 12.3 9; 135 6 12.3 9; 136 9 12.3 9; 137 12 12.3 9; 138 15 12.3 9
   34. 139 0 15.3 9; 140 3 15.3 9; 141 6 15.3 9; 142 9 15.3 9; 143 12 15.3 9
   35. 144 15 15.3 9; 145 0 0 12; 146 3 0 12; 147 6 0 12; 148 9 0 12
   36. 149 12 0 12; 150 15 0 12; 151 0 3.3 12; 152 3 3.3 12; 153 6 3.3 12
   37. 154 9 3.3 12; 155 12 3.3 12; 156 15 3.3 12; 157 0 6.3 12; 158 3 6.3 12
   38. 159 6 6.3 12; 160 9 6.3 12; 161 12 6.3 12; 162 15 6.3 12; 163 0 9.3 12
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39. 164 3 9.3 12; 165 6 9.3 12; 166 9 9.3 12; 167 12 9.3 12; 168 15 9.3 12
40. 169 0 12.3 12; 170 3 12.3 12; 171 6 12.3 12; 172 9 12.3 12
41. 173 12 12.3 12; 174 15 12.3 12; 175 0 15.3 12; 176 3 15.3 12
42. 177 6 15.3 12; 178 9 15.3 12; 179 12 15.3 12; 180 15 15.3 12
43. MEMBER INCIDENCES
44. 1 7 8; 2 8 9; 3 9 10; 4 10 11; 5 11 12; 6 13 14; 7 14 15; 8 15 16
45. 9 16 17; 10 17 18; 11 19 20; 12 20 21; 13 21 22; 14 22 23; 15 23 24
46. 16 25 26; 17 26 27; 18 27 28; 19 28 29; 20 29 30; 21 31 32; 22 32 33
47. 23 33 34; 24 34 35; 25 35 36; 26 1 7; 27 2 8; 28 3 9; 29 4 10; 30 5 11
48. 31 6 12; 32 7 13; 33 8 14; 34 9 15; 35 10 16; 36 11 17; 37 12 18
49. 38 13 19; 39 14 20; 40 15 21; 41 16 22; 42 17 23; 43 18 24; 44 19 25
50. 45 20 26; 46 21 27; 47 22 28; 48 23 29; 49 24 30; 50 25 31; 51 26 32
51. 52 27 33; 53 28 34; 54 29 35; 55 30 36; 56 43 44; 57 44 45; 58 45 46
52. 59 46 47; 60 47 48; 61 49 50; 62 50 51; 63 51 52; 64 52 53; 65 53 54
53. 66 55 56; 67 56 57; 68 57 58; 69 58 59; 70 59 60; 71 61 62; 72 62 63
54. 73 63 64; 74 64 65; 75 65 66; 76 67 68; 77 68 69; 78 69 70; 79 70 71
55. 80 71 72; 81 37 43; 82 38 44; 83 39 45; 84 40 46; 85 41 47; 86 42 48
56. 87 43 49; 88 44 50; 89 45 51; 90 46 52; 91 47 53; 92 48 54; 93 49 55
57. 94 50 56; 95 51 57; 96 52 58; 97 53 59; 98 54 60; 99 55 61; 100 56 62
58. 101 57 63; 102 58 64; 103 59 65; 104 60 66; 105 61 67; 106 62 68
59. 107 63 69; 108 64 70; 109 65 71; 110 66 72; 111 79 80; 112 80 81
60. 113 81 82; 114 82 83; 115 83 84; 116 85 86; 117 86 87; 118 87 88
61. 119 88 89; 120 89 90; 121 91 92; 122 92 93; 123 93 94; 124 94 95
62. 125 95 96; 126 97 98; 127 98 99; 128 99 100; 129 100 101; 130 101 102
63. 131 103 104; 132 104 105; 133 105 106; 134 106 107; 135 107 108
64. 136 73 79; 137 74 80; 138 75 81; 139 76 82; 140 77 83; 141 78 84
65. 142 79 85; 143 80 86; 144 81 87; 145 82 88; 146 83 89; 147 84 90
66. 148 85 91; 149 86 92; 150 87 93; 151 88 94; 152 89 95; 153 90 96
67. 154 91 97; 155 92 98; 156 93 99; 157 94 100; 158 95 101; 159 96 102
68. 160 97 103; 161 98 104; 162 99 105; 163 100 106; 164 101 107
69. 165 102 108; 166 115 116; 167 116 117; 168 117 118; 169 118 119
70. 170 119 120; 171 121 122; 172 122 123; 173 123 124; 174 124 125
71. 175 125 126; 176 127 128; 177 128 129; 178 129 130; 179 130 131
72. 180 131 132; 181 133 134; 182 134 135; 183 135 136; 184 136 137
73. 185 137 138; 186 139 140; 187 140 141; 188 141 142; 189 142 143
74. 190 143 144; 191 109 115; 192 110 116; 193 111 117; 194 112 118
75. 195 113 119; 196 114 120; 197 115 121; 198 116 122; 199 117 123
76. 200 118 124; 201 119 125; 202 120 126; 203 121 127; 204 122 128
77. 205 123 129; 206 124 130; 207 125 131; 208 126 132; 209 127 133
78. 210 128 134; 211 129 135; 212 130 136; 213 131 137; 214 132 138
79. 215 133 139; 216 134 140; 217 135 141; 218 136 142; 219 137 143
80. 220 138 144; 221 151 152; 222 152 153; 223 153 154; 224 154 155
81. 225 155 156; 226 157 158; 227 158 159; 228 159 160; 229 160 161
82. 230 161 162; 231 163 164; 232 164 165; 233 165 166; 234 166 167
83. 235 167 168; 236 169 170; 237 170 171; 238 171 172; 239 172 173
84. 240 173 174; 241 175 176; 242 176 177; 243 177 178; 244 178 179
85. 245 179 180; 246 145 151; 247 146 152; 248 147 153; 249 148 154
86. 250 149 155; 251 150 156; 252 151 157; 253 152 158; 254 153 159
87. 255 154 160; 256 155 161; 257 156 162; 258 157 163; 259 158 164
88. 260 159 165; 261 160 166; 262 161 167; 263 162 168; 264 163 169
89. 265 164 170; 266 165 171; 267 166 172; 268 167 173; 269 168 174
90. 270 169 175; 271 170 176; 272 171 177; 273 172 178; 274 173 179
91. 275 174 180; 276 7 43; 277 8 44; 278 9 45; 279 10 46; 280 11 47
92. 281 12 48; 282 13 49; 283 14 50; 284 15 51; 285 16 52; 286 17 53
93. 287 18 54; 288 19 55; 289 20 56; 290 21 57; 291 22 58; 292 23 59
94. 293 24 60; 294 25 61; 295 26 62; 296 27 63; 297 28 64; 298 29 65

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 95. 299 30 66; 300 31 67; 301 32 68; 302 33 69; 303 34 70; 304 35 71
 96. 305 36 72; 306 43 79; 307 44 80; 308 45 81; 309 46 82; 310 47 83
 97. 311 48 84; 312 49 85; 313 50 86; 314 51 87; 315 52 88; 316 53 89
 98. 317 54 90; 318 55 91; 319 56 92; 320 57 93; 321 58 94; 322 59 95
 99. 323 60 96; 324 61 97; 325 62 98; 326 63 99; 327 64 100; 328 65 101
100. 329 66 102; 330 67 103; 331 68 104; 332 69 105; 333 70 106; 334 71 107
101. 335 72 108; 336 79 115; 337 80 116; 338 81 117; 339 82 118; 340 83 119
102. 341 84 120; 342 85 121; 343 86 122; 344 87 123; 345 88 124; 346 89 125
103. 347 90 126; 348 91 127; 349 92 128; 350 93 129; 351 94 130; 352 95 131
104. 353 96 132; 354 97 133; 355 98 134; 356 99 135; 357 100 136
105. 358 101 137; 359 102 138; 360 103 139; 361 104 140; 362 105 141
106. 363 106 142; 364 107 143; 365 108 144; 366 115 151; 367 116 152
107. 368 117 153; 369 118 154; 370 119 155; 371 120 156; 372 121 157
108. 373 122 158; 374 123 159; 375 124 160; 376 125 161; 377 126 162
109. 378 127 163; 379 128 164; 380 129 165; 381 130 166; 382 131 167
110. 383 132 168; 384 133 169; 385 134 170; 386 135 171; 387 136 172
111. 388 137 173; 389 138 174; 390 139 175; 391 140 176; 392 141 177
112. 393 142 178; 394 143 179; 395 144 180
113. ELEMENT INCIDENCES SHELL
114. 396 7 8 44 43; 397 8 9 45 44; 398 9 10 46 45; 399 10 11 47 46
115. 400 11 12 48 47; 401 43 44 80 79; 402 44 45 81 80; 403 45 46 82 81
116. 404 46 47 83 82; 405 47 48 84 83; 406 79 80 116 115; 407 80 81 117 116
117. 408 81 82 118 117; 409 82 83 119 118; 410 83 84 120 119
118. 411 115 116 152 151; 412 116 117 153 152; 413 117 118 154 153
119. 414 118 119 155 154; 415 119 120 156 155; 416 13 14 50 49
120. 417 14 15 51 50; 418 15 16 52 51; 419 16 17 53 52; 420 17 18 54 53
121. 421 49 50 86 85; 422 50 51 87 86; 423 51 52 88 87; 424 52 53 89 88
122. 425 53 54 90 89; 426 85 86 122 121; 427 86 87 123 122
123. 428 87 88 124 123; 429 88 89 125 124; 430 89 90 126 125
124. 431 121 122 158 157; 432 122 123 159 158; 433 123 124 160 159
125. 434 124 125 161 160; 435 125 126 162 161; 436 19 20 56 55
126. 437 20 21 57 56; 438 21 22 58 57; 439 22 23 59 58; 440 23 24 60 59
127. 441 55 56 92 91; 442 56 57 93 92; 443 57 58 94 93; 444 58 59 95 94
128. 445 59 60 96 95; 446 91 92 128 127; 447 92 93 129 128
129. 448 93 94 130 129; 449 94 95 131 130; 450 95 96 132 131
130. 451 127 128 164 163; 452 128 129 165 164; 453 129 130 166 165
131. 454 130 131 167 166; 455 131 132 168 167; 456 25 26 62 61
132. 457 26 27 63 62; 458 27 28 64 63; 459 28 29 65 64; 460 29 30 66 65
133. 461 61 62 98 97; 462 62 63 99 98; 463 63 64 100 99; 464 64 65 101 100
134. 465 65 66 102 101; 466 97 98 134 133; 467 98 99 135 134
135. 468 99 100 136 135; 469 100 101 137 136; 470 101 102 138 137
136. 471 133 134 170 169; 472 134 135 171 170; 473 135 136 172 171
137. 474 136 137 173 172; 475 137 138 174 173; 476 31 32 68 67
138. 477 32 33 69 68; 478 33 34 70 69; 479 34 35 71 70; 480 35 36 72 71
139. 481 67 68 104 103; 482 68 69 105 104; 483 69 70 106 105
140. 484 70 71 107 106; 485 71 72 108 107; 486 103 104 140 139
141. 487 104 105 141 140; 488 105 106 142 141; 489 106 107 143 142
142. 490 107 108 144 143; 491 139 140 176 175; 492 140 141 177 176
143. 493 141 142 178 177; 494 142 143 179 178; 495 143 144 180 179
144. ELEMENT PROPERTY
145. 396 TO 495 THICKNESS 0.15
146. DEFINE MATERIAL START
147. ISOTROPIC CONCRETE
148. E 2.17185E+007
149. POISSON 0.17
150. DENSITY 23.5616
```

```
  151. ALPHA 1E-005
  152. DAMP 0.05
  153. TYPE CONCRETE
  154. STRENGTH FCU 27579
  155. END DEFINE MATERIAL
  156. MEMBER PROPERTY
  157. 26 TO 55 81 TO 110 136 TO 165 191 TO 220 246 TO 275 PRIS YD 0.6 ZD 0.4
  158. 1 TO 25 56 TO 80 111 TO 135 166 TO 190 221 TO 245 276 TO 394 -
  159. 395 PRIS YD 0.4 ZD 0.4
  160. CONSTANTS
  161. MATERIAL CONCRETE ALL
  162. SUPPORTS
  163. 1 TO 6 37 TO 42 73 TO 78 109 TO 114 145 TO 150 FIXED
  164. DEFINE 1893 LOAD
  165. ZONE 0.16 RF 5 I 1 SS 2 ST 1 DM 0.05
  166. SELFWEIGHT 1
  167. MEMBER WEIGHT
  168. 1 TO 25 56 TO 80 111 TO 135 166 TO 190 221 TO 245 276 TO 395 UNI 24.3
  169. FLOOR WEIGHT
**NOTE** about Floor/OneWay Loads/Weights.
  Please note that depending on the shape of the floor you may
  have to break up the FLOOR/ONEWAY LOAD into multiple commands.
  For details please refer to Technical Reference Manual
  Section 5.32.4.2 Note d and/or "5.32.4.3 Note f.

  170. YRANGE 3 15 FLOAD 2
  171. DEFINE WIND LOAD

  *** NOTE: If any floor diaphragm is present in the model Wind Load definition
            should be defined after Floor Diaphragm definition. Otherwise wind
            load generation may be unsuccessful during analysis.

  172. TYPE 1 WIND 1
  173. <! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
  174. ASCE-7-2010:PARAMS 44.000 M/SEC 0 1 2 0 0.000 FT 0.000 FT 0.000 FT 1 -
  175. 1 15.000 M 15.000 M 12.000 M 2.000 0.010 0 -
  176. 0 0 0 0 0.807 1.000 1.000 0.850 0 -
  177. 0 0 0 0.855 0.800 -0.550
  178. !> END GENERATED DATA BLOCK
  179. INT 0.844871 0.844871 0.863631 0.880485 0.89584 0.909981 0.923116 -
  180. 0.935401 0.946958 0.957881 0.968249 0.978125 0.98756 0.996601 1.00528 -
  181. HEIG  0  4.572  5.37415  6.17631  6.97846  7.78061  8.58277  9.38492 -
  182. 10.1871  10.9892  11.7914  12.5935  13.3957  14.1979  15
  183. EXP 1 JOINT 1 TO 180
  184. LOAD 1 LOADTYPE SEISMIC  TITLE EQX
  185. 1893 LOAD X 1
  186. LOAD 2 LOADTYPE SEISMIC  TITLE EQY
  187. 1893 LOAD Z 1
  188. LOAD 3 LOADTYPE WIND  TITLE WX
  189. WIND LOAD X 1 TYPE 1
```

```
190. WIND LOAD X -1 TYPE 1
191. LOAD 4 LOADTYPE WIND  TITLE WY
192. WIND LOAD Z 1 TYPE 1
193. WIND LOAD Z -1 TYPE 1
194. LOAD 5 LOADTYPE DEAD  TITLE DL
195. SELFWEIGHT Y -1 LIST ALL
196. MEMBER LOAD
197. 1 TO 275 UNI GY -24.3
198. FLOOR LOAD
199. YRANGE 3 15 FLOAD -2 GY
200. LOAD 6 LOADTYPE LIVE REDUCIBLE TITLE LL
201. FLOOR LOAD
202. YRANGE 3 15 FLOAD -3 GY
203. LOAD COMB 7 GENERATED INDIAN CODE GENRAL_STRUCTURES 1
204. 5 1.5 6 1.5
205. LOAD COMB 8 GENERATED INDIAN CODE GENRAL_STRUCTURES 2
206. 5 1.2 6 1.2 3 1.2
207. LOAD COMB 9 GENERATED INDIAN CODE GENRAL_STRUCTURES 3
208. 5 1.2 6 1.2 4 1.2
209. LOAD COMB 10 GENERATED INDIAN CODE GENRAL_STRUCTURES 4
210. 5 1.2 6 1.2 3 -1.2
211. LOAD COMB 11 GENERATED INDIAN CODE GENRAL_STRUCTURES 5
212. 5 1.2 6 1.2 4 -1.2
213. LOAD COMB 12 GENERATED INDIAN CODE GENRAL_STRUCTURES 6
214. 5 1.2 6 1.2 1 1.2
215. LOAD COMB 13 GENERATED INDIAN CODE GENRAL_STRUCTURES 7
216. 5 1.2 6 1.2 2 1.2
217. LOAD COMB 14 GENERATED INDIAN CODE GENRAL_STRUCTURES 8
218. 5 1.2 6 1.2 1 -1.2
219. LOAD COMB 15 GENERATED INDIAN CODE GENRAL_STRUCTURES 9
220. 5 1.2 6 1.2 2 -1.2
221. LOAD COMB 16 GENERATED INDIAN CODE GENRAL_STRUCTURES 10
222. 5 1.5 3 1.5
223. LOAD COMB 17 GENERATED INDIAN CODE GENRAL_STRUCTURES 11
224. 5 1.5 4 1.5
225. LOAD COMB 18 GENERATED INDIAN CODE GENRAL_STRUCTURES 12
226. 5 1.5 3 -1.5
227. LOAD COMB 19 GENERATED INDIAN CODE GENRAL_STRUCTURES 13
```

```
228. 5 1.5 4 -1.5
229. LOAD COMB 20 GENERATED INDIAN CODE GENRAL_STRUCTURES 14
230. 5 1.5 1 1.5
231. LOAD COMB 21 GENERATED INDIAN CODE GENRAL_STRUCTURES 15
232. 5 1.5 2 1.5
233. LOAD COMB 22 GENERATED INDIAN CODE GENRAL_STRUCTURES 16
234. 5 1.5 1 -1.5
235. LOAD COMB 23 GENERATED INDIAN CODE GENRAL_STRUCTURES 17
236. 5 1.5 2 -1.5
237. LOAD COMB 24 GENERATED INDIAN CODE GENRAL_STRUCTURES 18
238. 5 0.9 1 1.5
239. LOAD COMB 25 GENERATED INDIAN CODE GENRAL_STRUCTURES 19
240. 5 0.9 2 1.5
241. LOAD COMB 26 GENERATED INDIAN CODE GENRAL_STRUCTURES 20
242. 5 0.9 1 -1.5
243. LOAD COMB 27 GENERATED INDIAN CODE GENRAL_STRUCTURES 21
244. 5 0.9 2 -1.5
245. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**180**	**NUMBER OF MEMBERS**	**395**
NUMBER OF PLATES	**100**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**30**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 37/ 34/ 204 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 900
TOTAL LOAD COMBINATION CASES = 21 SO FAR.
SIZE OF STIFFNESS MATRIX = 184 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 15.2/ 45674.5 MB

```
**WARNING: IF THIS UBC/IBC ANALYSIS HAS TENSION/COMPRESSION
OR REPEAT LOAD OR RE-ANALYSIS OR SELECT OPTIMIZE, THEN EACH
UBC/IBC CASE SHOULD BE FOLLOWED BY PERFORM ANALYSIS _CHANGE.
```

```
************************************************************
* UNITS - KN    METE                                       *
* TIME PERIOD FOR X 1893 LOADING =    0.58020 SEC          *
* SA/G PER 1893=    2.344, LOAD FACTOR= 1.000              *
* VB PER 1893=    0.0375 X    27847.70=    1044.40 KN      *
*                                                          *
************************************************************
```

```
*********************************************************
* UNITS - KN    METE                                    *
* TIME PERIOD FOR Z 1893 LOADING =    0.58020 SEC       *
* SA/G PER 1893=    2.344, LOAD FACTOR= 1.000           *
* VB PER 1893=    0.0375 X    27847.70=    1044.40 KN   *
*                                                       *
*********************************************************
```

246. START CONCRETE DESIGN

```
247. CODE INDIAN
248. FC 30000 ALL
249. FYMAIN 415000 ALL
250. FYSEC 415000 ALL
251. MAXSEC 12 ALL
252. MAXMAIN 20 ALL
253. DESIGN BEAM 1 TO 25 56 TO 80 111 TO 135 166 TO 190 221 TO 245 276 TO 395
```

IS 456 - 2000 B E A M D E S I G N R E S U L T S

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 1 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	786.64 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	703.41 (Sq. mm)
BOTTOM REINF.	316.84 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	7-12í 1 layer(s)
BOTTOM REINF.	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 78.67 MX = 0.25 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -75.21 MX = -0.44 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 2 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	717.87 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	715.10 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 75.07 MX = -0.03 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -74.73 MX = -0.02 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 3 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	719.01 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	719.01 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 75.08 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -75.01 MX = -0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 4 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	715.10 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	717.87 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 74.80 MX = 0.02 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -75.00 MX = 0.03 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 5 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	703.41 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	786.64 (Sq. mm)
BOTTOM REINF.	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	312.46 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	7-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 75.28 MX = 0.44 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -78.59 MX = -0.25 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 6 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	831.50 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	673.49 (Sq. mm)
BOTTOM REINF.	320.38 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	328.95 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	11-10í 2 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.7 mm AWAY FROM START SUPPORT
VY = 82.23 MX = 0.22 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -73.06 MX = -0.36 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 7 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	750.82 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	746.32 (Sq. mm)
BOTTOM REINF.	306.08 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	304.34 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 77.62 MX = -0.04 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -77.28 MX = -0.06 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 8 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	750.15 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	750.15 (Sq. mm)
BOTTOM REINF.	305.61 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 77.48 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -77.41 MX = -0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 9 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	746.32 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	750.82 (Sq. mm)
BOTTOM REINF.	305.32 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.81 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 77.35 MX = 0.06 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -77.55 MX = 0.04 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 10 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	673.49 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	831.50 (Sq. mm)
BOTTOM REINF.	328.95 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	11-10í 2 layer(s)
BOTTOM REINF.	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 73.13 MX = 0.36 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 667.7 mm AWAY FROM END SUPPORT
VY = -82.16 MX = -0.22 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 11 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	770.49 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	578.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	8-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 78.26 MX = 0.19 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -65.56 MX = -0.32 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 12 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	691.90 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	683.35 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 73.11 MX = -0.04 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -72.48 MX = -0.08 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 13 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	690.04 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	690.04 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 72.86 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -72.79 MX = -0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 14 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	683.35 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	691.90 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 72.55 MX = 0.08 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -73.04 MX = 0.04 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 15 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	578.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	770.49 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	8-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 65.63 MX = 0.32 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -78.19 MX = -0.19 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 16 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	649.76 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	432.39 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	6-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 69.12 MX = 0.09 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -53.18 MX = -0.23 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 17 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	579.10 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	570.27 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM START SUPPORT
VY = 63.85 MX = -0.05 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM END SUPPORT
VY = -63.25 MX = -0.10 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 18 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	576.92 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	576.92 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM START SUPPORT
VY = 63.60 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM END SUPPORT
VY = -63.53 MX = -0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 19 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	570.27 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	579.10 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM START SUPPORT
VY = 63.32 MX = 0.10 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM END SUPPORT
VY = -63.78 MX = 0.05 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 20 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	432.39 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	649.76 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	6-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 53.25 MX = 0.23 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -69.05 MX = -0.09 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 21 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	432.83 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 51.35 MX = 0.12 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -41.83 MX = -0.19 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 22 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	396.65 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	399.49 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 48.25 MX = -0.06 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -48.80 MX = -0.15 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 23 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	399.46 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	399.46 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 48.29 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -48.29 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 24 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	399.49 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	396.65 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 48.80 MX = 0.15 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -48.25 MX = 0.06 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 25 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	432.83 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 41.83 MX = 0.19 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -51.35 MX = -0.12 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 56 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	768.29 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	670.43 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 79.12 MX = -0.05 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -74.93 MX = -0.01 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 57 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	700.08 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	697.03 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 75.03 MX = -0.01 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -74.58 MX = -0.01 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 58 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	701.92 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	701.92 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 75.05 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -74.91 MX = -0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 59 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	697.03 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	700.08 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 74.72 MX = 0.01 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -74.89 MX = 0.01 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 60 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	670.43 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	768.29 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 75.07 MX = 0.01 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -78.98 MX = 0.05 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 61 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	825.16 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	646.19 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	11-10í 2 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.7 mm AWAY FROM START SUPPORT
VY = 83.00 MX = -0.06 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -72.58 MX = -0.06 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 62 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	735.67 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	729.59 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 77.80 MX = -0.02 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -77.31 MX = -0.01 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 63 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	735.30 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	735.30 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 77.64 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -77.50 MX = -0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 64 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	729.59 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	735.67 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 77.45 MX = 0.01 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -77.66 MX = 0.02 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 65 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	646.19 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	825.16 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	11-10í 2 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 72.72 MX = 0.06 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 667.7 mm AWAY FROM END SUPPORT
VY = -82.86 MX = 0.06 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 66 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	768.73 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	553.30 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	8-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 79.34 MX = -0.08 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -65.18 MX = -0.09 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 67 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	680.69 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	669.96 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 73.59 MX = -0.02 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -72.79 MX = -0.02 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 68 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	678.99 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	678.99 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 73.30 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -73.16 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 69 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	669.96 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	680.69 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 72.93 MX = 0.02 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -73.45 MX = 0.02 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 70 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	553.30 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	768.73 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	8-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 65.32 MX = 0.09 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -79.20 MX = 0.08 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 71 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	656.39 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	415.30 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	6-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 70.82 MX = -0.08 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -53.25 MX = -0.11 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 72 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	576.92 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	564.53 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM START SUPPORT
VY = 65.04 MX = -0.02 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM END SUPPORT
VY = -64.17 MX = -0.03 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 73 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	574.33 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	574.33 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM START SUPPORT
VY = 64.70 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM END SUPPORT
VY = -64.55 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 74 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	564.53 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	576.92 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM START SUPPORT
VY = 64.31 MX = 0.03 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM END SUPPORT
VY = -64.90 MX = 0.02 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 75 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	415.30 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	656.39 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	6-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 53.39 MX = 0.11 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -70.68 MX = 0.08 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 76 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	427.51 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 50.78 MX = -0.11 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -39.57 MX = -0.16 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 77 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	380.45 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	380.87 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 47.10 MX = -0.03 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -47.61 MX = -0.04 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 78 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	383.66 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	383.66 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 47.10 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -47.10 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 79 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	380.87 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	380.45 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 47.61 MX = 0.04 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -47.10 MX = 0.03 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 80 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	427.51 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 39.57 MX = 0.16 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -50.78 MX = 0.11 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 111 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	768.58 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	669.89 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 79.18 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -74.90 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 112 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	700.25 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	697.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 75.06 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -74.59 MX = -0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 113 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	702.26 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	702.26 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 75.08 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -74.94 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 114 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	697.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	700.25 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 74.73 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -74.92 MX = -0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 115 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	669.89 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	768.58 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 75.04 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -79.04 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 116 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	825.65 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	644.55 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	11-10í 2 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.7 mm AWAY FROM START SUPPORT
VY = 83.09 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -72.51 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 117 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	735.79 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	729.32 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 77.83 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -77.31 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 118 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	735.46 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	735.46 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 77.65 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -77.51 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 119 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	729.32 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	735.79 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 77.45 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -77.69 MX = -0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 120 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	644.55 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	825.65 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	11-10í 2 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 72.65 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 667.7 mm AWAY FROM END SUPPORT
VY = -82.95 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 121 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	769.56 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	551.27 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	8-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 79.47 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -65.09 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 122 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	680.94 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	669.68 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 73.64 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -72.79 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 123 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	679.27 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	679.27 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 73.32 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -73.18 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 124 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	669.68 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	680.94 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 72.94 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -73.49 MX = -0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 125 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	551.27 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	769.56 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	8-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 65.23 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -79.33 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 126 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	657.69 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	412.44 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	6-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 70.99 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -53.10 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 127 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	577.28 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	563.95 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM START SUPPORT
VY = 65.09 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM END SUPPORT
VY = -64.15 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 128 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	574.50 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	574.50 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM START SUPPORT
VY = 64.71 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM END SUPPORT
VY = -64.57 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 129 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	563.95 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	577.28 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM START SUPPORT
VY = 64.29 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM END SUPPORT
VY = -64.95 MX = -0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 130 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	412.44 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	657.69 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	6-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 53.24 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -70.85 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 131 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	427.51 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 50.88 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -39.49 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 132 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	380.27 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	380.36 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 47.12 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -47.63 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 133 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	383.78 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	383.78 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 47.11 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -47.11 MX = -0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 134 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	380.36 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	380.27 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 47.63 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -47.12 MX = -0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 135 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	427.51 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c	2 legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 39.49 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 225 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -50.88 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 225 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 166 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	768.29 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	670.43 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 79.12 MX = 0.05 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -74.93 MX = 0.01 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 167 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	700.08 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	697.03 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 75.03 MX = 0.01 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -74.58 MX = 0.01 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 168 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	701.92 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	701.92 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 75.05 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -74.91 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 169 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	697.03 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	700.08 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 74.72 MX = -0.01 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -74.89 MX = -0.01 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 170 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	670.43 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	768.29 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 75.07 MX = -0.01 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -78.98 MX = -0.05 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 171 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	825.16 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	646.19 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	11-10í 2 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.7 mm AWAY FROM START SUPPORT
VY = 83.00 MX = 0.06 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -72.58 MX = 0.06 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 172 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	735.67 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	729.59 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 77.80 MX = 0.02 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -77.31 MX = 0.01 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 173 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	735.30 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	735.30 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 77.64 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -77.50 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 174 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	729.59 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	735.67 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 77.45 MX = -0.01 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -77.66 MX = -0.02 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 175 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	646.19 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	825.16 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	11-10í 2 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 72.72 MX = -0.06 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 667.7 mm AWAY FROM END SUPPORT
VY = -82.86 MX = -0.06 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 176 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	768.73 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	553.30 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	8-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 79.34 MX = 0.08 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -65.18 MX = 0.09 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 177 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	680.69 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	669.96 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 73.59 MX = 0.02 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -72.79 MX = 0.02 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 178 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	678.99 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	678.99 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 73.30 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -73.16 MX = -0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 179 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	669.96 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	680.69 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 72.93 MX = -0.02 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -73.45 MX = -0.02 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 180 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	553.30 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	768.73 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	8-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 65.32 MX = -0.09 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -79.20 MX = -0.08 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 181 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	656.39 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	415.30 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	6-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 70.82 MX = 0.08 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -53.25 MX = 0.11 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 182 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	576.92 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	564.53 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM START SUPPORT
VY = 65.04 MX = 0.02 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM END SUPPORT
VY = -64.17 MX = 0.03 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 183 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	574.33 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	574.33 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM START SUPPORT
VY = 64.70 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM END SUPPORT
VY = -64.55 MX = -0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 184 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	564.53 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	576.92 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM START SUPPORT
VY = 64.31 MX = -0.03 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM END SUPPORT
VY = -64.90 MX = -0.02 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 185 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	415.30 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	656.39 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	6-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 53.39 MX = -0.11 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -70.68 MX = -0.08 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 186 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	427.51 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 50.78 MX = 0.11 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -39.57 MX = 0.16 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 187 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	380.45 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	380.87 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 47.10 MX = 0.03 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -47.61 MX = 0.04 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 188 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	383.66 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	383.66 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 47.10 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -47.10 MX = -0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 189 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	380.87 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	380.45 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 47.61 MX = -0.04 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -47.10 MX = -0.03 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 190 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	427.51 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 39.57 MX = -0.16 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -50.78 MX = -0.11 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 221 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	786.64 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	703.41 (Sq. mm)
BOTTOM REINF.	316.84 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	7-12í 1 layer(s)
BOTTOM REINF.	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 78.67 MX = -0.25 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -75.21 MX = 0.44 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 222 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	717.87 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	715.10 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 75.07 MX = 0.03 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -74.73 MX = 0.02 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 223 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	719.01 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	719.01 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 75.08 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -75.01 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 224 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	715.10 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	717.87 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 74.80 MX = -0.02 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -75.00 MX = -0.03 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 225 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	703.41 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	786.64 (Sq. mm)
BOTTOM REINF.	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	312.46 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	7-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 75.28 MX = -0.44 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -78.59 MX = 0.25 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 226 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	831.50 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	673.49 (Sq. mm)
BOTTOM REINF.	320.38 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	328.95 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	11-10í 2 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.7 mm AWAY FROM START SUPPORT
VY = 82.23 MX = -0.22 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -73.06 MX = 0.36 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 227 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	750.82 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	746.32 (Sq. mm)
BOTTOM REINF.	306.08 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	304.34 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 77.62 MX = 0.04 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -77.28 MX = 0.06 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 228 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	750.15 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	750.15 (Sq. mm)
BOTTOM REINF.	305.61 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 77.48 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -77.41 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 229 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	746.32 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	750.82 (Sq. mm)
BOTTOM REINF.	305.32 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.81 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 77.35 MX = -0.06 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -77.55 MX = -0.04 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 230 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	673.49 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	831.50 (Sq. mm)
BOTTOM REINF.	328.95 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	11-10í 2 layer(s)
BOTTOM REINF.	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 73.13 MX = -0.36 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 667.7 mm AWAY FROM END SUPPORT
VY = -82.16 MX = 0.22 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 231 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	770.49 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	578.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	10-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	8-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 78.26 MX = -0.19 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -65.56 MX = 0.32 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 232 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	691.90 (Sq. mm)	303.13 (Sq. mm)	0.00 (Sq. mm)	303.13 (Sq. mm)	683.35 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 73.11 MX = 0.04 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -72.48 MX = 0.08 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 233 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	690.04 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	690.04 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 72.86 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -72.79 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 234 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	683.35 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	691.90 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	9-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	9-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 72.55 MX = -0.08 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -73.04 MX = -0.04 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 235 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	578.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	770.49 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	8-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	10-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM START SUPPORT
VY = 65.63 MX = -0.32 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 670.0 mm AWAY FROM END SUPPORT
VY = -78.19 MX = 0.19 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 236 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	649.76 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	432.39 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	6-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 69.12 MX = -0.09 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -53.18 MX = 0.23 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 237 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	579.10 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	570.27 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM START SUPPORT
VY = 63.85 MX = 0.05 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM END SUPPORT
VY = -63.25 MX = 0.10 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 238 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	576.92 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	576.92 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM START SUPPORT
VY = 63.60 MX = -0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM END SUPPORT
VY = -63.53 MX = 0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 239 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	570.27 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	579.10 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM START SUPPORT
VY = 63.32 MX = -0.10 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 667.0 mm AWAY FROM END SUPPORT
VY = -63.78 MX = -0.05 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 240 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	432.39 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	649.76 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	6-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 53.25 MX = -0.23 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -69.05 MX = 0.09 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 241 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	432.83 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 51.35 MX = -0.12 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -41.83 MX = 0.19 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 242 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	396.65 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	399.49 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 48.25 MX = 0.06 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -48.80 MX = 0.15 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 243 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	399.46 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	399.46 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 48.29 MX = 0.00 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -48.29 MX = -0.00 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 244 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	399.49 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	396.65 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 48.80 MX = -0.15 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -48.25 MX = -0.06 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 245 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	432.83 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM START SUPPORT
VY = 41.83 MX = -0.19 LD= 22
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 669.0 mm AWAY FROM END SUPPORT
VY = -51.35 MX = 0.12 LD= 20
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 276 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	657.54 (Sq. mm)	308.87 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	563.34 (Sq. mm)
BOTTOM REINF.	573.42 (Sq. mm)	322.80 (Sq. mm)	300.67 (Sq. mm)	300.67 (Sq. mm)	494.22 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	6-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	5-12í 1 layer(s)
BOTTOM REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 55.49 MX = -0.17 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -54.59 MX = 0.31 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 277 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	622.67 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	515.19 (Sq. mm)
BOTTOM REINF.	500.44 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	422.49 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	8-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 54.17 MX = 0.04 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -52.64 MX = 0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 278 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	622.82 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	514.78 (Sq. mm)
BOTTOM REINF.	500.35 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	423.04 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	8-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 54.24 MX = 0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -52.63 MX = 0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 279 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	622.82 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	514.78 (Sq. mm)
BOTTOM REINF.	500.35 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	423.04 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	8-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 54.24 MX = -0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -52.63 MX = -0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 280 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	622.67 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	515.19 (Sq. mm)
BOTTOM REINF.	500.44 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	422.49 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	8-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 54.17 MX = -0.04 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -52.64 MX = -0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 281 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	657.54 (Sq. mm)	308.87 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	563.34 (Sq. mm)
BOTTOM REINF.	573.42 (Sq. mm)	322.80 (Sq. mm)	300.67 (Sq. mm)	300.67 (Sq. mm)	494.22 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	6-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	5-12í 1 layer(s)
BOTTOM REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 55.49 MX = 0.17 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -54.59 MX = -0.31 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 282 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	630.45 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	532.59 (Sq. mm)
BOTTOM REINF.	524.59 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	475.00 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	6-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	5-12í 1 layer(s)
BOTTOM REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 54.05 MX = -0.10 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -51.49 MX = 0.26 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 283 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	606.42 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	486.04 (Sq. mm)
BOTTOM REINF.	461.70 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	416.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	8-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 53.56 MX = 0.06 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -49.68 MX = 0.07 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 284 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	606.32 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	484.71 (Sq. mm)
BOTTOM REINF.	461.30 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	416.79 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	8-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 53.63 MX = 0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -49.65 MX = 0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 285 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	606.32 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	484.71 (Sq. mm)
BOTTOM REINF.	461.30 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	416.79 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	8-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 53.63 MX = -0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -49.65 MX = -0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 286 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	606.42 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	486.04 (Sq. mm)
BOTTOM REINF.	461.70 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	416.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	8-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 53.56 MX = -0.06 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -49.68 MX = -0.07 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 287 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	630.45 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	532.59 (Sq. mm)
BOTTOM REINF.	524.59 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	475.00 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	6-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	5-12í 1 layer(s)
BOTTOM REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 54.05 MX = 0.10 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -51.49 MX = -0.26 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 288 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	548.64 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	446.20 (Sq. mm)
BOTTOM REINF.	437.86 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	400.47 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
BOTTOM REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 47.86 MX = -0.05 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -44.36 MX = 0.26 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 289 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	533.40 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	404.21 (Sq. mm)
BOTTOM REINF.	378.07 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	348.43 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 47.98 MX = 0.08 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -42.77 MX = 0.09 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 290 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	533.56 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	402.43 (Sq. mm)
BOTTOM REINF.	377.60 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	349.38 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 48.09 MX = 0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -42.72 MX = 0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 291 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	533.56 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	402.43 (Sq. mm)
BOTTOM REINF.	377.60 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	349.38 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 48.09 MX = -0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -42.72 MX = -0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 292 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	533.40 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	404.21 (Sq. mm)
BOTTOM REINF.	378.07 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	348.43 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 47.98 MX = -0.08 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -42.77 MX = -0.09 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 293 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	548.64 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	446.20 (Sq. mm)
BOTTOM REINF.	437.86 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	400.47 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
BOTTOM REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 47.86 MX = 0.05 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -44.36 MX = -0.26 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 294 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	394.61 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.39 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 36.19 MX = 0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -31.82 MX = 0.21 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 295 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	394.03 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 37.40 MX = 0.08 LD= 15
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -31.08 MX = 0.10 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 296 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	394.23 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 37.53 MX = 0.01 LD= 15
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -30.99 MX = 0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 297 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	394.23 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 37.53 MX = -0.01 LD= 15
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -30.99 MX = -0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 298 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	394.03 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 37.40 MX = -0.08 LD= 15
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -31.08 MX = -0.10 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 299 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	394.61 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.39 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 36.19 MX = -0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -31.82 MX = -0.21 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 300 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 17.76 MX = -0.03 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -14.87 MX = 0.15 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 301 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 17.10 MX = 0.11 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -12.37 MX = 0.15 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 302 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 17.13 MX = 0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -12.38 MX = 0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 303 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 17.13 MX = -0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -12.38 MX = -0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 304 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 17.10 MX = -0.11 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -12.37 MX = -0.15 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 305 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 17.76 MX = 0.03 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -14.87 MX = -0.15 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 306 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	531.38 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	537.52 (Sq. mm)
BOTTOM REINF.	448.45 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	454.69 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 49.15 MX = 0.03 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -49.06 MX = -0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 307 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	490.33 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	497.84 (Sq. mm)
BOTTOM REINF.	382.61 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	389.76 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 47.42 MX = 0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -47.25 MX = 0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 308 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	490.90 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	498.55 (Sq. mm)
BOTTOM REINF.	383.36 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	390.59 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 47.48 MX = 0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -47.31 MX = 0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 309 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	490.90 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	498.55 (Sq. mm)
BOTTOM REINF.	383.36 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	390.59 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 47.48 MX = -0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -47.31 MX = -0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 310 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	490.33 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	497.84 (Sq. mm)
BOTTOM REINF.	382.61 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	389.76 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 47.42 MX = -0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -47.25 MX = -0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 311 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	531.38 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	537.52 (Sq. mm)
BOTTOM REINF.	448.45 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	454.69 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 49.15 MX = -0.03 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -49.06 MX = 0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 312 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	543.26 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	544.37 (Sq. mm)
BOTTOM REINF.	460.64 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	459.99 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 49.79 MX = 0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -49.79 MX = 0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 313 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	506.32 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	508.30 (Sq. mm)
BOTTOM REINF.	401.35 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	400.47 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 48.50 MX = 0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -48.47 MX = 0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 314 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	506.61 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	508.60 (Sq. mm)
BOTTOM REINF.	401.93 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	401.05 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 48.55 MX = 0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -48.51 MX = 0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 315 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	506.61 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	508.60 (Sq. mm)
BOTTOM REINF.	401.93 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	401.05 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 48.55 MX = -0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -48.51 MX = -0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 316 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	506.32 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	508.30 (Sq. mm)
BOTTOM REINF.	401.35 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	400.47 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 48.50 MX = -0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -48.47 MX = -0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 317 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	543.26 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	544.37 (Sq. mm)
BOTTOM REINF.	460.64 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	459.99 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 49.79 MX = -0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -49.79 MX = -0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 318 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	477.87 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	477.85 (Sq. mm)
BOTTOM REINF.	399.85 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	398.02 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 44.65 MX = 0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -44.65 MX = 0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 319 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	449.65 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	449.91 (Sq. mm)
BOTTOM REINF.	345.71 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	343.25 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 43.90 MX = 0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -43.86 MX = 0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 320 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	450.02 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	450.24 (Sq. mm)
BOTTOM REINF.	346.46 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	343.98 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 43.95 MX = 0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -43.91 MX = 0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 321 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	450.02 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	450.24 (Sq. mm)
BOTTOM REINF.	346.46 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	343.98 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 43.95 MX = -0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -43.91 MX = -0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 322 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	449.65 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	449.91 (Sq. mm)
BOTTOM REINF.	345.71 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	343.25 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 43.90 MX = -0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -43.86 MX = -0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 323 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	477.87 (Sq. mm)	303.13 (Sq. mm)	0.00 (Sq. mm)	303.13 (Sq. mm)	477.85 (Sq. mm)
BOTTOM REINF.	399.85 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	398.02 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 44.65 MX = -0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -44.65 MX = -0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 324 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	360.92 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	358.10 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 34.98 MX = 0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -34.95 MX = 0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 325 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	346.08 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	342.61 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 35.27 MX = 0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -35.17 MX = 0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 326 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	346.38 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	342.68 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 35.32 MX = 0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -35.21 MX = 0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 327 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	346.38 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	342.68 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 35.32 MX = -0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -35.21 MX = -0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 328 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	346.08 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	342.61 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 35.27 MX = -0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -35.17 MX = -0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 329 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	360.92 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	358.10 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 34.98 MX = -0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -34.95 MX = -0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 330 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 17.43 MX = 0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -17.94 MX = 0.06 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 331 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 15.97 MX = 0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -16.68 MX = 0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 332 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 15.97 MX = 0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -16.73 MX = 0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 333 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 15.97 MX = -0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -16.73 MX = -0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 334 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 15.97 MX = -0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -16.68 MX = -0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 335 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 17.43 MX = -0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -17.94 MX = -0.06 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 336 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	537.52 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	531.38 (Sq. mm)
BOTTOM REINF.	454.69 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	448.45 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 49.15 MX = 0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -49.06 MX = -0.03 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 337 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	497.84 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	490.33 (Sq. mm)
BOTTOM REINF.	389.76 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	382.61 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 47.43 MX = -0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -47.23 MX = -0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 338 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	498.55 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	490.90 (Sq. mm)
BOTTOM REINF.	390.59 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	383.36 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 47.49 MX = -0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -47.30 MX = -0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 339 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	498.55 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	490.90 (Sq. mm)
BOTTOM REINF.	390.59 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	383.36 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 47.49 MX = 0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -47.30 MX = 0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 340 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	497.84 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	490.33 (Sq. mm)
BOTTOM REINF.	389.76 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	382.61 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 47.43 MX = 0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -47.23 MX = 0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 341 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	537.52 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	531.38 (Sq. mm)
BOTTOM REINF.	454.69 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	448.45 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 49.15 MX = -0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -49.06 MX = 0.03 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 342 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	544.37 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	543.26 (Sq. mm)
BOTTOM REINF.	459.99 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	460.64 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 49.89 MX = -0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -49.70 MX = -0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 343 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	508.30 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	506.32 (Sq. mm)
BOTTOM REINF.	400.47 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	401.35 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 48.65 MX = -0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -48.32 MX = -0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 344 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	508.60 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	506.61 (Sq. mm)
BOTTOM REINF.	401.05 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	401.93 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 48.69 MX = -0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -48.37 MX = -0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 345 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	508.60 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	506.61 (Sq. mm)
BOTTOM REINF.	401.05 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	401.93 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 48.69 MX = 0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -48.37 MX = 0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 346 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	508.30 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	506.32 (Sq. mm)
BOTTOM REINF.	400.47 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	401.35 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 48.65 MX = 0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -48.32 MX = 0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 347 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	544.37 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	543.26 (Sq. mm)
BOTTOM REINF.	459.99 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	460.64 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 49.89 MX = 0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -49.70 MX = 0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 348 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	477.85 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	477.87 (Sq. mm)
BOTTOM REINF.	398.02 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	399.85 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 44.75 MX = -0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -44.55 MX = -0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 349 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	449.91 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	449.65 (Sq. mm)
BOTTOM REINF.	343.25 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	345.71 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 44.04 MX = -0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -43.71 MX = -0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 350 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	450.24 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	450.02 (Sq. mm)
BOTTOM REINF.	343.98 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	346.46 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 44.09 MX = -0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -43.77 MX = -0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 351 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	450.24 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	450.02 (Sq. mm)
BOTTOM REINF.	343.98 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	346.46 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 44.09 MX = 0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -43.77 MX = 0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 352 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	449.91 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	449.65 (Sq. mm)
BOTTOM REINF.	343.25 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	345.71 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 44.04 MX = 0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -43.71 MX = 0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 353 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	477.85 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	477.87 (Sq. mm)
BOTTOM REINF.	398.02 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	399.85 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 44.75 MX = 0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -44.55 MX = 0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 354 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	358.10 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	360.92 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 35.04 MX = -0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -34.89 MX = -0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 355 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	342.61 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	346.08 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 35.36 MX = -0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -35.09 MX = -0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 356 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	342.68 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	346.38 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 35.39 MX = -0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -35.14 MX = -0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 357 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	342.68 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	346.38 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 35.39 MX = 0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -35.14 MX = 0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 358 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	342.61 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	346.08 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 35.36 MX = 0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -35.09 MX = 0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 359 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	358.10 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	360.92 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 35.04 MX = 0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -34.89 MX = 0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 360 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 17.94 MX = -0.06 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -17.43 MX = -0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 361 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 16.68 MX = -0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -15.97 MX = -0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 362 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 16.73 MX = -0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -15.97 MX = -0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 363 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 16.73 MX = 0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -15.97 MX = 0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 364 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 16.68 MX = 0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -15.97 MX = 0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 365 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 17.94 MX = 0.06 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -17.43 MX = 0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 366 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	563.34 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	308.87 (Sq. mm)	657.54 (Sq. mm)
BOTTOM REINF.	494.22 (Sq. mm)	300.67 (Sq. mm)	300.67 (Sq. mm)	322.80 (Sq. mm)	573.42 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	6-12í 1 layer(s)
BOTTOM REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 54.68 MX = -0.31 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -55.40 MX = 0.17 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 367 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	515.19 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	622.67 (Sq. mm)
BOTTOM REINF.	422.49 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	500.44 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	8-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 52.83 MX = -0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -53.99 MX = -0.04 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 368 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	514.78 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	622.82 (Sq. mm)
BOTTOM REINF.	423.04 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	500.35 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	8-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 52.81 MX = -0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -54.05 MX = -0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 369 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	514.78 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	622.82 (Sq. mm)
BOTTOM REINF.	423.04 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	500.35 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	8-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 52.81 MX = 0.00 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -54.05 MX = 0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 370 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	515.19 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	622.67 (Sq. mm)
BOTTOM REINF.	422.49 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	500.44 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	8-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 52.83 MX = 0.02 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -53.99 MX = 0.04 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 371 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	563.34 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	308.87 (Sq. mm)	657.54 (Sq. mm)
BOTTOM REINF.	494.22 (Sq. mm)	300.67 (Sq. mm)	300.67 (Sq. mm)	322.80 (Sq. mm)	573.42 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	6-12í 1 layer(s)
BOTTOM REINF.	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)	3-16í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 54.68 MX = 0.31 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -55.40 MX = -0.17 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 372 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	532.59 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	630.45 (Sq. mm)
BOTTOM REINF.	475.00 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	524.59 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	6-12í 1 layer(s)
BOTTOM REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 51.58 MX = -0.26 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -53.95 MX = 0.10 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 373 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	486.04 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	606.42 (Sq. mm)
BOTTOM REINF.	416.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	461.70 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	8-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 49.86 MX = -0.07 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -53.37 MX = -0.06 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 374 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	484.71 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	606.32 (Sq. mm)
BOTTOM REINF.	416.79 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	461.30 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	8-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 49.83 MX = -0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -53.45 MX = -0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 375 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	484.71 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	606.32 (Sq. mm)
BOTTOM REINF.	416.79 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	461.30 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	8-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 49.83 MX = 0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -53.45 MX = 0.00 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 376 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	486.04 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	606.42 (Sq. mm)
BOTTOM REINF.	416.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	461.70 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	8-10í 1 layer(s)
BOTTOM REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	6-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 49.86 MX = 0.07 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -53.37 MX = 0.06 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 377 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	532.59 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	630.45 (Sq. mm)
BOTTOM REINF.	475.00 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	524.59 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	5-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	6-12í 1 layer(s)
BOTTOM REINF.	7-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 51.58 MX = 0.26 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -53.95 MX = -0.10 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 378 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	446.20 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	548.64 (Sq. mm)
BOTTOM REINF.	400.47 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	437.86 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 44.45 MX = -0.26 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -47.77 MX = 0.05 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 379 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	404.21 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	533.40 (Sq. mm)
BOTTOM REINF.	348.43 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	378.07 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 42.95 MX = -0.09 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -47.79 MX = -0.08 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 380 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	402.43 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	533.56 (Sq. mm)
BOTTOM REINF.	349.38 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	377.60 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 42.90 MX = -0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -47.91 MX = -0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 381 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	402.43 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	533.56 (Sq. mm)
BOTTOM REINF.	349.38 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	377.60 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 42.90 MX = 0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -47.91 MX = 0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 382 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	404.21 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	533.40 (Sq. mm)
BOTTOM REINF.	348.43 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	378.07 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	5-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	5-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 42.95 MX = 0.09 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -47.79 MX = 0.08 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 383 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	446.20 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	548.64 (Sq. mm)
BOTTOM REINF.	400.47 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	302.31 (Sq. mm)	437.86 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	6-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	7-10í 1 layer(s)
BOTTOM REINF.	4-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 44.45 MX = 0.26 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -47.77 MX = -0.05 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 384 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.39 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	394.61 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 31.91 MX = -0.21 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -36.10 MX = -0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 385 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	394.03 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 31.33 MX = -0.10 LD= 15
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -37.17 MX = -0.08 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 386 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	394.23 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 31.23 MX = -0.01 LD= 15
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -37.30 MX = -0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 387 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	394.23 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 31.23 MX = 0.01 LD= 15
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -37.30 MX = 0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 388 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	394.03 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 31.33 MX = 0.10 LD= 15
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -37.17 MX = 0.08 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 389 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.39 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	394.61 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	3-12í 1 layer(s)	4-12í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM START SUPPORT
VY = 31.91 MX = 0.21 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 569.0 mm AWAY FROM END SUPPORT
VY = -36.10 MX = 0.02 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 390 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 14.87 MX = -0.15 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -17.76 MX = 0.03 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 391 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 12.37 MX = -0.15 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -17.10 MX = -0.11 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 392 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 12.38 MX = -0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -17.13 MX = -0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 393 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 12.38 MX = 0.01 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -17.13 MX = 0.01 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 394 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 12.37 MX = 0.15 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -17.10 MX = 0.11 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

IS-456 L I M I T S T A T E D E S I G N
B E A M N O. 395 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm SIZE: 400.0 mm X 400.0 mm COVER: 25.0 mm

SUMMARY OF REINF. AREA (Sq.mm)

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)
BOTTOM REINF.	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)	303.13 (Sq. mm)

SUMMARY OF PROVIDED REINF. AREA

SECTION	0.0 mm	750.0 mm	1500.0 mm	2250.0 mm	3000.0 mm
TOP REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
BOTTOM REINF.	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)	4-10í 1 layer(s)
SHEAR REINF.	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c	2 legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT DISTANCE d (EFFECTIVE DEPTH) FROM FACE OF THE SUPPORT

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM START SUPPORT
VY = 14.87 MX = 0.15 LD= 23
Provide 2 Legged 8í @ 175 mm c/c

SHEAR DESIGN RESULTS AT 570.0 mm AWAY FROM END SUPPORT
VY = -17.76 MX = -0.03 LD= 21
Provide 2 Legged 8í @ 175 mm c/c

**********END OF BEAM DESIGN RESULTS**********

254. DESIGN COLUMN 26 TO 55 81 TO 110 136 TO 165 191 TO 220 246 TO 275

IS 456 - 2000 C O L U M N D E S I G N R E S U L T S

IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 26 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3300.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 1 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 149.29 Muy1 : 95.44

INTERACTION RATIO: 0.50 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 2
END JOINT: 1 Puz : 3913.49 Muz : 176.06 Muy : 111.81 IR: 0.57

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     27  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:     2  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.87   Muy1 :    114.47

INTERACTION RATIO: 0.44 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:     2
END JOINT:     2 Puz :   3913.49   Muz :    171.75   Muy :    109.16   IR: 0.61
================================================================================


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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     28  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:     3  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 140.83 Muy1 : 90.20

INTERACTION RATIO: 0.73 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 2
END JOINT: 3 Puz : 3913.49 Muz : 171.85 Muy : 109.22 IR: 0.61
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 29 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3300.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 4 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.50 Muy1 : 114.85

INTERACTION RATIO: 0.44 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 2
END JOINT: 4 Puz : 3913.49 Muz : 171.85 Muy : 109.22 IR: 0.61

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     30  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:     5  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    140.72   Muy1 :     90.13

INTERACTION RATIO: 0.73 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:     2
END JOINT:     5 Puz :   3913.49   Muz :    171.75   Muy :    109.16   IR: 0.61
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     31  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:     6  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 145.13 Muy1 : 92.85

INTERACTION RATIO: 0.69 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 2
END JOINT: 6 Puz : 3913.49 Muz : 176.06 Muy : 111.81 IR: 0.57
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 32 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 7 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 157.62 Muy1 : 100.62

INTERACTION RATIO: 0.22 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 13 Puz : 3913.49 Muz : 321.12 Muy : 205.10 IR: 0.30

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     33  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:     8  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.45   Muy1 :    114.21

INTERACTION RATIO: 0.32 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:     2
END JOINT:    14 Puz :   3913.49   Muz :    183.08   Muy :    116.13   IR: 0.33
================================================================================


================================================================================
            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     34  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:     9  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 152.26 Muy1 : 97.29

INTERACTION RATIO: 0.39 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 2
END JOINT: 15 Puz : 3913.49 Muz : 183.06 Muy : 116.12 IR: 0.33
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 35 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 10 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.47 Muy1 : 114.83

INTERACTION RATIO: 0.32 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 2
END JOINT: 16 Puz : 3913.49 Muz : 183.06 Muy : 116.12 IR: 0.33

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     36  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    11  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    152.28   Muy1 :     97.31

INTERACTION RATIO: 0.39 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:     2
END JOINT:    17 Puz :   3913.49   Muz :    183.08   Muy :    116.13   IR: 0.33
================================================================================


================================================================================
            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     37  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    12  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 155.21 Muy1 : 99.13

INTERACTION RATIO: 0.34 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 18 Puz : 3913.49 Muz : 321.12 Muy : 205.10 IR: 0.30
==========

==========
IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 38 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 13 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 166.08 Muy1 : 105.85

INTERACTION RATIO: 0.17 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 23
END JOINT: 19 Puz : 3913.49 Muz : 324.83 Muy : 207.47 IR: 0.34

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================================================================================
            IS-456    L I M I T    S T A T E    D E S I G N
        C O L U M N   N O.     39  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:    14  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.41   Muy1 :    114.18

INTERACTION RATIO: 0.27 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    27
END JOINT:    20 Puz :   3913.49   Muz :    311.96   Muy :    199.06   IR: 0.33
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            IS-456    L I M I T    S T A T E    D E S I G N
        C O L U M N   N O.     40  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    15  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 163.07 Muy1 : 103.99

INTERACTION RATIO: 0.32 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 21 Puz : 3913.49 Muz : 313.99 Muy : 200.38 IR: 0.33
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 41 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 16 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.47 Muy1 : 114.83

INTERACTION RATIO: 0.27 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 22 Puz : 3913.49 Muz : 313.99 Muy : 200.38 IR: 0.33

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     42  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    17  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    163.15   Muy1 :    104.04

INTERACTION RATIO: 0.32 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    27
END JOINT:    23 Puz :   3913.49   Muz :    311.96   Muy :    199.06   IR: 0.33
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     43  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    18  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 164.74 Muy1 : 105.02

INTERACTION RATIO: 0.29 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 23
END JOINT: 24 Puz : 3913.49 Muz : 324.83 Muy : 207.47 IR: 0.34
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 44 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 19 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 173.23 Muy1 : 110.31

INTERACTION RATIO: 0.11 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 23
END JOINT: 25 Puz : 3913.49 Muz : 284.41 Muy : 181.16 IR: 0.35

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     45  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:    20  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.60   Muy1 :    114.30

INTERACTION RATIO: 0.20 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    27
END JOINT:    26 Puz :   3913.49   Muz :    275.39   Muy :    175.27   IR: 0.31
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     46  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    21  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 172.08 Muy1 : 109.59

INTERACTION RATIO: 0.22 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 27 Puz : 3913.49 Muz : 277.32 Muy : 176.50 IR: 0.31
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 47 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 22 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.46 Muy1 : 114.83

INTERACTION RATIO: 0.20 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 28 Puz : 3913.49 Muz : 277.32 Muy : 176.50 IR: 0.31

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     48  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    23  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    172.17   Muy1 :    109.65

INTERACTION RATIO: 0.22 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    27
END JOINT:    29 Puz :   3913.49   Muz :    275.39   Muy :    175.27   IR: 0.31
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     49  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    24  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 172.75 Muy1 : 110.01

INTERACTION RATIO: 0.20 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 23
END JOINT: 30 Puz : 3913.49 Muz : 284.41 Muy : 181.16 IR: 0.35
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 50 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 25 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 178.12 Muy1 : 113.37

INTERACTION RATIO: 0.03 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 23
END JOINT: 31 Puz : 3913.49 Muz : 233.46 Muy : 147.53 IR: 0.39

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     51  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:    26  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.11   Muy1 :    114.61

INTERACTION RATIO: 0.10 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    22
END JOINT:    32 Puz :   3913.49   Muz :    243.76   Muy :    154.33   IR: 0.29
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     52  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    27  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 178.07 Muy1 : 113.34

INTERACTION RATIO: 0.09 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 22
END JOINT: 33 Puz : 3913.49 Muz : 244.60 Muy : 154.88 IR: 0.27
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 53 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 28 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.46 Muy1 : 114.83

INTERACTION RATIO: 0.11 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 20
END JOINT: 34 Puz : 3913.49 Muz : 244.60 Muy : 154.88 IR: 0.27

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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 54 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 2 END JOINT: 29 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 178.13 Muy1 : 113.38

INTERACTION RATIO: 0.09 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 20
END JOINT: 35 Puz : 3913.49 Muz : 243.76 Muy : 154.33 IR: 0.29

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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 55 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 2 END JOINT: 30 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 178.18 Muy1 : 113.41

INTERACTION RATIO: 0.08 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 23
END JOINT: 36 Puz : 3913.49 Muz : 233.46 Muy : 147.53 IR: 0.39
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 81 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3300.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 37 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 144.19 Muy1 : 92.28

INTERACTION RATIO: 0.54 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 37 Puz : 3913.49 Muz : 336.82 Muy : 215.30 IR: 0.55

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     82  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:    38  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.64   Muy1 :    114.32

INTERACTION RATIO: 0.46 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:     2
END JOINT:    38 Puz :   3913.49   Muz :    212.45   Muy :    134.29   IR: 0.54
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     83  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:    39  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.50 Muy1 : 114.86

INTERACTION RATIO: 0.46 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 2
END JOINT: 39 Puz : 3913.49 Muz : 212.48 Muy : 134.31 IR: 0.54
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 84 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3300.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 3 END JOINT: 40 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.51 Muy1 : 114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 40 Puz : 3913.49 Muz : 358.58 Muy : 229.32 IR: 0.50

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     85  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    41  SHORT COLUMN


REQD. STEEL AREA   :     1178.31 Sq.mm.
REQD. CONCRETE AREA:   147288.61 Sq.mm.
MAIN REINFORCEMENT : Provide  12 - 12 dia. (0.57%,   1357.17 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3590.84   Muz1 :    115.91   Muy1 :     75.01

INTERACTION RATIO: 0.97 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:     2
END JOINT:    41 Puz :   3644.10   Muz :    131.95   Muy :     84.86   IR: 0.86
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     86  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    3 END JOINT:    42  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.51 Muy1 : 114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 42 Puz : 3913.49 Muz : 336.82 Muy : 215.30 IR: 0.55
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 87 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 43 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 154.14 Muy1 : 98.46

INTERACTION RATIO: 0.25 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 49 Puz : 3913.49 Muz : 307.94 Muy : 196.42 IR: 0.49

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            IS-456    L I M I T    S T A T E    D E S I G N
        C O L U M N   N O.     88  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:    44  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.09   Muy1 :    113.98

INTERACTION RATIO: 0.37 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    27
END JOINT:    50 Puz :   3913.49   Muz :    330.65   Muy :    211.27   IR: 0.52
================================================================================


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            IS-456    L I M I T    S T A T E    D E S I G N
        C O L U M N   N O.     89  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    3 END JOINT:    45  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.50 Muy1 : 114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 51 Puz : 3913.49 Muz : 333.33 Muy : 213.01 IR: 0.51
==========

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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 90 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 46 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.47 Muy1 : 114.83

INTERACTION RATIO: 0.36 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 52 Puz : 3913.49 Muz : 333.33 Muy : 213.01 IR: 0.51

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     91  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    3 END JOINT:    47  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.50   Muy1 :    114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    27
END JOINT:    53 Puz :   3913.49   Muz :    330.65   Muy :    211.27   IR: 0.52
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================================================================================
            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     92  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    4 END JOINT:    48  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.50 Muy1 : 114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 54 Puz : 3913.49 Muz : 307.94 Muy : 196.42 IR: 0.49
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 93 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 49 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 164.12 Muy1 : 104.64

INTERACTION RATIO: 0.19 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 55 Puz : 3913.49 Muz : 284.99 Muy : 181.55 IR: 0.50

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     94  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:    50  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.05   Muy1 :    113.95

INTERACTION RATIO: 0.31 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    27
END JOINT:    56 Puz :   3913.49   Muz :    305.04   Muy :    194.48   IR: 0.51
================================================================================


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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     95  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    51  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.21 Muy1 : 114.68

INTERACTION RATIO: 0.50 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 57 Puz : 3913.49 Muz : 307.64 Muy : 196.22 IR: 0.51
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 96 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 52 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.46 Muy1 : 114.83

INTERACTION RATIO: 0.31 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 58 Puz : 3913.49 Muz : 307.64 Muy : 196.22 IR: 0.51

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     97  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    53  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.20   Muy1 :    114.67

INTERACTION RATIO: 0.50 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    27
END JOINT:    59 Puz :   3913.49   Muz :    305.04   Muy :    194.48   IR: 0.51
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.     98  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    54  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.34 Muy1 : 114.75

INTERACTION RATIO: 0.45 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 60 Puz : 3913.49 Muz : 284.99 Muy : 181.55 IR: 0.50
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 99 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 55 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 172.45 Muy1 : 109.82

INTERACTION RATIO: 0.11 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 61 Puz : 3913.49 Muz : 257.14 Muy : 163.13 IR: 0.48

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    100  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:    56  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.29   Muy1 :    114.11

INTERACTION RATIO: 0.23 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    27
END JOINT:    62 Puz :   3913.49   Muz :    272.68   Muy :    173.52   IR: 0.46
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    101  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    57  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 179.83 Muy1 : 114.44

INTERACTION RATIO: 0.39 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 63 Puz : 3913.49 Muz : 275.02 Muy : 175.08 IR: 0.46
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 102 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 58 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.45 Muy1 : 114.82

INTERACTION RATIO: 0.24 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 64 Puz : 3913.49 Muz : 275.02 Muy : 175.08 IR: 0.46

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    103  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    59  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.83   Muy1 :    114.44

INTERACTION RATIO: 0.39 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    27
END JOINT:    65 Puz :   3913.49   Muz :    272.68   Muy :    173.52   IR: 0.46
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    104  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    60  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.03 Muy1 : 114.56

INTERACTION RATIO: 0.35 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 66 Puz : 3913.49 Muz : 257.14 Muy : 163.13 IR: 0.48
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 105 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 61 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 177.97 Muy1 : 113.28

INTERACTION RATIO: 0.03 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 23
END JOINT: 67 Puz : 3913.49 Muz : 234.68 Muy : 148.32 IR: 0.49

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    106  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:    62  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.95   Muy1 :    114.51

INTERACTION RATIO: 0.12 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    23
END JOINT:    68 Puz :   3913.49   Muz :    248.03   Muy :    157.15   IR: 0.38
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    107  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    63  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.10 Muy1 : 114.61

INTERACTION RATIO: 0.22 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 69 Puz : 3913.49 Muz : 235.17 Muy : 148.64 IR: 0.35
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 108 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 64 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.45 Muy1 : 114.82

INTERACTION RATIO: 0.13 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 70 Puz : 3913.49 Muz : 235.17 Muy : 148.64 IR: 0.35

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    109  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    65  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.10   Muy1 :    114.60

INTERACTION RATIO: 0.22 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    23
END JOINT:    71 Puz :   3913.49   Muz :    248.03   Muy :    157.15   IR: 0.38
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    110  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    66  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.23 Muy1 : 114.69

INTERACTION RATIO: 0.19 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 23
END JOINT: 72 Puz : 3913.49 Muz : 234.68 Muy : 148.32 IR: 0.49
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 136 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3300.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 73 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 144.34 Muy1 : 92.37

INTERACTION RATIO: 0.54 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 73 Puz : 3913.49 Muz : 338.70 Muy : 216.50 IR: 0.55

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    137  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:    74  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.66   Muy1 :    114.34

INTERACTION RATIO: 0.46 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:     2
END JOINT:    74 Puz :   3913.49   Muz :    210.52   Muy :    133.09   IR: 0.54
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    138  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:    75  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.51 Muy1 : 114.86

INTERACTION RATIO: 0.46 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 2
END JOINT: 75 Puz : 3913.49 Muz : 210.52 Muy : 133.09 IR: 0.54
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 139 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3300.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 2 END JOINT: 76 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.51 Muy1 : 114.86

INTERACTION RATIO: 0.63 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 2
END JOINT: 76 Puz : 3913.49 Muz : 210.52 Muy : 133.09 IR: 0.54

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    140  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    77  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.51   Muy1 :    114.86

INTERACTION RATIO: 0.63 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:     2
END JOINT:    77 Puz :   3913.49   Muz :    210.52   Muy :    133.09   IR: 0.54
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================================================================================
            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    141  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    78  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.51 Muy1 : 114.86

INTERACTION RATIO: 0.61 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 78 Puz : 3913.49 Muz : 338.70 Muy : 216.50 IR: 0.55
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 142 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 79 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 154.13 Muy1 : 98.46

INTERACTION RATIO: 0.25 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 85 Puz : 3913.49 Muz : 309.23 Muy : 197.27 IR: 0.48

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    143  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:    80  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.09   Muy1 :    113.98

INTERACTION RATIO: 0.37 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    27
END JOINT:    86 Puz :   3913.49   Muz :    332.16   Muy :    212.25   IR: 0.50
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    144  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    81  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.51 Muy1 : 114.86

INTERACTION RATIO: 0.55 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 87 Puz : 3913.49 Muz : 334.91 Muy : 214.05 IR: 0.50
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 145 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 82 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.46 Muy1 : 114.83

INTERACTION RATIO: 0.36 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 88 Puz : 3913.49 Muz : 334.91 Muy : 214.05 IR: 0.50

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    146  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    83  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.51   Muy1 :    114.86

INTERACTION RATIO: 0.55 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    27
END JOINT:    89 Puz :   3913.49   Muz :    332.16   Muy :    212.25   IR: 0.50
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    147  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    84  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.51 Muy1 : 114.86

INTERACTION RATIO: 0.49 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 90 Puz : 3913.49 Muz : 309.23 Muy : 197.27 IR: 0.48
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 148 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 85 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 164.03 Muy1 : 104.59

INTERACTION RATIO: 0.19 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 91 Puz : 3913.49 Muz : 285.74 Muy : 182.04 IR: 0.50

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    149  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:    86  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.03   Muy1 :    113.95

INTERACTION RATIO: 0.31 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    25
END JOINT:    92 Puz :   3913.49   Muz :    305.93   Muy :    195.10   IR: 0.50
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    150  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    87  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.51 Muy1 : 114.86

INTERACTION RATIO: 0.50 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 93 Puz : 3913.49 Muz : 308.62 Muy : 196.87 IR: 0.50
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 151 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 88 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.46 Muy1 : 114.83

INTERACTION RATIO: 0.31 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 94 Puz : 3913.49 Muz : 308.62 Muy : 196.87 IR: 0.50

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    152  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    89  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.51   Muy1 :    114.86

INTERACTION RATIO: 0.50 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    27
END JOINT:    95 Puz :   3913.49   Muz :    305.93   Muy :    195.10   IR: 0.50
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    153  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    90  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.51 Muy1 : 114.86

INTERACTION RATIO: 0.45 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 96 Puz : 3913.49 Muz : 285.74 Muy : 182.04 IR: 0.50
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 154 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 91 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 172.34 Muy1 : 109.76

INTERACTION RATIO: 0.11 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 97 Puz : 3913.49 Muz : 257.32 Muy : 163.25 IR: 0.48

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    155  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:    92  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.27   Muy1 :    114.10

INTERACTION RATIO: 0.23 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    25
END JOINT:    98 Puz :   3913.49   Muz :    272.99   Muy :    173.72   IR: 0.46
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    156  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    93  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.51 Muy1 : 114.86

INTERACTION RATIO: 0.40 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 99 Puz : 3913.49 Muz : 275.50 Muy : 175.33 IR: 0.46
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 157 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 94 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.45 Muy1 : 114.82

INTERACTION RATIO: 0.24 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 100 Puz : 3913.49 Muz : 275.50 Muy : 175.33 IR: 0.46

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    158  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    95  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.51   Muy1 :    114.86

INTERACTION RATIO: 0.40 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    25
END JOINT:   101 Puz :   3913.49   Muz :    272.99   Muy :    173.72   IR: 0.46
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    159  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    96  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.51 Muy1 : 114.86

INTERACTION RATIO: 0.35 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 102 Puz : 3913.49 Muz : 257.32 Muy : 163.25 IR: 0.48
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 160 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 97 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 177.91 Muy1 : 113.24

INTERACTION RATIO: 0.03 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 23
END JOINT: 103 Puz : 3913.49 Muz : 234.98 Muy : 148.51 IR: 0.50

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    161  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:    98  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.94   Muy1 :    114.51

INTERACTION RATIO: 0.12 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    21
END JOINT:   104 Puz :   3913.49   Muz :    248.53   Muy :    157.48   IR: 0.38
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    162  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:    99  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.51 Muy1 : 114.86

INTERACTION RATIO: 0.23 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 105 Puz : 3913.49 Muz : 235.33 Muy : 148.74 IR: 0.36
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 163 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 100 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.44 Muy1 : 114.82

INTERACTION RATIO: 0.13 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 106 Puz : 3913.49 Muz : 235.33 Muy : 148.74 IR: 0.36

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    164  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:   101  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.51   Muy1 :    114.86

INTERACTION RATIO: 0.23 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    21
END JOINT:   107 Puz :   3913.49   Muz :    248.53   Muy :    157.48   IR: 0.38
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    165  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:   102  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.51 Muy1 : 114.86

INTERACTION RATIO: 0.20 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 21
END JOINT: 108 Puz : 3913.49 Muz : 234.98 Muy : 148.51 IR: 0.50
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 191 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3300.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 109 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 144.19 Muy1 : 92.28

INTERACTION RATIO: 0.54 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 109 Puz : 3913.49 Muz : 336.82 Muy : 215.30 IR: 0.55

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    192  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:   110  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.64   Muy1 :    114.32

INTERACTION RATIO: 0.46 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:     2
END JOINT:   110 Puz :   3913.49   Muz :    208.57   Muy :    131.88   IR: 0.55
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    193  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:   111  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.50 Muy1 : 114.86

INTERACTION RATIO: 0.46 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 2
END JOINT: 111 Puz : 3913.49 Muz : 208.54 Muy : 131.86 IR: 0.55
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 194 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3300.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 2 END JOINT: 112 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 178.46 Muy1 : 113.59

INTERACTION RATIO: 0.64 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 2
END JOINT: 112 Puz : 3913.49 Muz : 208.54 Muy : 131.86 IR: 0.55

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    195  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:   113  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    178.49   Muy1 :    113.60

INTERACTION RATIO: 0.64 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:     2
END JOINT:   113 Puz :   3913.49   Muz :    208.57   Muy :    131.88   IR: 0.55
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    196  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:   114  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 178.82 Muy1 : 113.81

INTERACTION RATIO: 0.62 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 114 Puz : 3913.49 Muz : 336.82 Muy : 215.30 IR: 0.55
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 197 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 115 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 154.14 Muy1 : 98.46

INTERACTION RATIO: 0.25 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 121 Puz : 3913.49 Muz : 307.94 Muy : 196.42 IR: 0.49

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    198  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:   116  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.09   Muy1 :    113.98

INTERACTION RATIO: 0.37 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    25
END JOINT:   122 Puz :   3913.49   Muz :    330.65   Muy :    211.27   IR: 0.52
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    199  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:   117  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.01 Muy1 : 114.55

INTERACTION RATIO: 0.57 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 123 Puz : 3913.49 Muz : 333.33 Muy : 213.01 IR: 0.51
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 200 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 118 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.47 Muy1 : 114.83

INTERACTION RATIO: 0.36 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 124 Puz : 3913.49 Muz : 333.33 Muy : 213.01 IR: 0.51

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    201  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:   119  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.03   Muy1 :    114.56

INTERACTION RATIO: 0.57 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    25
END JOINT:   125 Puz :   3913.49   Muz :    330.65   Muy :    211.27   IR: 0.52
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    202  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:   120  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.05 Muy1 : 114.58

INTERACTION RATIO: 0.51 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 126 Puz : 3913.49 Muz : 307.94 Muy : 196.42 IR: 0.49
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 203 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 121 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 164.12 Muy1 : 104.64

INTERACTION RATIO: 0.19 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 127 Puz : 3913.49 Muz : 284.99 Muy : 181.55 IR: 0.50

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    204  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:   122  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.05   Muy1 :    113.95

INTERACTION RATIO: 0.31 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    25
END JOINT:   128 Puz :   3913.49   Muz :    305.04   Muy :    194.48   IR: 0.51
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    205  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    3 END JOINT:   123  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.50 Muy1 : 114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 129 Puz : 3913.49 Muz : 307.64 Muy : 196.22 IR: 0.51
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 206 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 124 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.46 Muy1 : 114.83

INTERACTION RATIO: 0.31 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 130 Puz : 3913.49 Muz : 307.64 Muy : 196.22 IR: 0.51

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    207  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    3 END JOINT:   125  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.50   Muy1 :    114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    25
END JOINT:   131 Puz :   3913.49   Muz :    305.04   Muy :    194.48   IR: 0.51
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    208  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    4 END JOINT:   126  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.50 Muy1 : 114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 132 Puz : 3913.49 Muz : 284.99 Muy : 181.55 IR: 0.50
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 209 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 127 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 172.45 Muy1 : 109.82

INTERACTION RATIO: 0.11 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 133 Puz : 3913.49 Muz : 257.14 Muy : 163.13 IR: 0.48

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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 210 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 128 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 179.29 Muy1 : 114.11

INTERACTION RATIO: 0.23 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 134 Puz : 3913.49 Muz : 272.68 Muy : 173.52 IR: 0.46

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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 211 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 3 END JOINT: 129 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.50 Muy1 : 114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 135 Puz : 3913.49 Muz : 275.02 Muy : 175.08 IR: 0.46
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 212 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 130 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.45 Muy1 : 114.82

INTERACTION RATIO: 0.24 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 136 Puz : 3913.49 Muz : 275.02 Muy : 175.08 IR: 0.46

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    213  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    3 END JOINT:   131  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.50   Muy1 :    114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    25
END JOINT:   137 Puz :   3913.49   Muz :    272.68   Muy :    173.52   IR: 0.46
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    214  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    4 END JOINT:   132  TENSION COLUMN
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REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.50 Muy1 : 114.85

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 138 Puz : 3913.49 Muz : 257.14 Muy : 163.13 IR: 0.48
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 215 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 133 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 177.97 Muy1 : 113.28

INTERACTION RATIO: 0.03 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 21
END JOINT: 139 Puz : 3913.49 Muz : 234.68 Muy : 148.32 IR: 0.49

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    216  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:   134  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.95   Muy1 :    114.51

INTERACTION RATIO: 0.12 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    21
END JOINT:   140 Puz :   3913.49   Muz :    248.03   Muy :    157.15   IR: 0.38
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    217  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    3 END JOINT:   135  TENSION COLUMN
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REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.50 Muy1 : 114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 141 Puz : 3913.49 Muz : 235.17 Muy : 148.64 IR: 0.35
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 218 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 136 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.45 Muy1 : 114.82

INTERACTION RATIO: 0.13 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 142 Puz : 3913.49 Muz : 235.17 Muy : 148.64 IR: 0.35

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    219  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    3 END JOINT:   137  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.50   Muy1 :    114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    21
END JOINT:   143 Puz :   3913.49   Muz :    248.03   Muy :    157.15   IR: 0.38
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    220  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    4 END JOINT:   138  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.50 Muy1 : 114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 21
END JOINT: 144 Puz : 3913.49 Muz : 234.68 Muy : 148.32 IR: 0.49
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 246 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3300.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 145 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 149.29 Muy1 : 95.44

INTERACTION RATIO: 0.50 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 145 Puz : 3913.49 Muz : 301.94 Muy : 192.53 IR: 0.52

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    247  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:   146  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.87   Muy1 :    114.47

INTERACTION RATIO: 0.44 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    27
END JOINT:   146 Puz :   3913.49   Muz :    325.67   Muy :    208.00   IR: 0.51
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    248  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    3 END JOINT:   147  TENSION COLUMN
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REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.51 Muy1 : 114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 147 Puz : 3913.49 Muz : 328.09 Muy : 209.58 IR: 0.51
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 249 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3300.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 148 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.50 Muy1 : 114.85

INTERACTION RATIO: 0.44 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 148 Puz : 3913.49 Muz : 328.09 Muy : 209.58 IR: 0.51

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    250  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    4 END JOINT:   149  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.51   Muy1 :    114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    27
END JOINT:   149 Puz :   3913.49   Muz :    325.67   Muy :    208.00   IR: 0.51
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    251  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3300.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    3 END JOINT:   150  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.51 Muy1 : 114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 150 Puz : 3913.49 Muz : 301.94 Muy : 192.53 IR: 0.52
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 252 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 151 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 157.62 Muy1 : 100.62

INTERACTION RATIO: 0.22 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 157 Puz : 3913.49 Muz : 321.12 Muy : 205.10 IR: 0.30

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    253  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:   152  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.45   Muy1 :    114.21

INTERACTION RATIO: 0.32 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    26
END JOINT:   152 Puz :   3913.49   Muz :    332.25   Muy :    212.31   IR: 0.33
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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    254  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    3 END JOINT:   153  TENSION COLUMN
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REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.50 Muy1 : 114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 26
END JOINT: 153 Puz : 3913.49 Muz : 333.54 Muy : 213.15 IR: 0.33
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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 255 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 154 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.47 Muy1 : 114.83

INTERACTION RATIO: 0.32 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 24
END JOINT: 154 Puz : 3913.49 Muz : 333.54 Muy : 213.15 IR: 0.33

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    256  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    3 END JOINT:   155  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.50   Muy1 :    114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    24
END JOINT:   155 Puz :   3913.49   Muz :    332.25   Muy :    212.31   IR: 0.33
================================================================================


================================================================================
            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    257  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:   21 END JOINT:   156  SHORT COLUMN
```

REQD. STEEL AREA : 650.77 Sq.mm.
REQD. CONCRETE AREA: 81346.62 Sq.mm.
MAIN REINFORCEMENT : Provide 8 - 12 dia. (0.38%, 904.78 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3433.77 Muz1 : 256.56 Muy1 : 168.44

INTERACTION RATIO: 0.31 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 162 Puz : 3509.40 Muz : 167.42 Muy : 107.77 IR: 0.48
==========

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IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 258 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 157 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 166.08 Muy1 : 105.85

INTERACTION RATIO: 0.17 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 21
END JOINT: 163 Puz : 3913.49 Muz : 324.83 Muy : 207.47 IR: 0.34

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================================================================================
            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    259  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:   158  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.41   Muy1 :    114.18

INTERACTION RATIO: 0.27 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    25
END JOINT:   164 Puz :   3913.49   Muz :    311.96   Muy :    199.06   IR: 0.33
================================================================================


================================================================================
            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    260  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    3 END JOINT:   159  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.50 Muy1 : 114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 165 Puz : 3913.49 Muz : 313.99 Muy : 200.38 IR: 0.33
==========

==========
IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 261 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 160 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.47 Muy1 : 114.83

INTERACTION RATIO: 0.27 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 166 Puz : 3913.49 Muz : 313.99 Muy : 200.38 IR: 0.33

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================================================================================


================================================================================
            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    262  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    3 END JOINT:   161  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.50   Muy1 :    114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    25
END JOINT:   167 Puz :   3913.49   Muz :    311.96   Muy :    199.06   IR: 0.33
================================================================================


================================================================================
            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    263  D E S I G N   R E S U L T S

      M30                   Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:   21 END JOINT:   162  SHORT COLUMN
```

REQD. STEEL AREA : 477.19 Sq.mm.
REQD. CONCRETE AREA: 59649.29 Sq.mm.
MAIN REINFORCEMENT : Provide 8 - 12 dia. (0.38%, 904.78 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3382.08 Muz1 : 213.53 Muy1 : 140.61

INTERACTION RATIO: 0.42 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 27
END JOINT: 168 Puz : 3509.40 Muz : 148.49 Muy : 95.71 IR: 0.53
==========

==========
IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 264 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 163 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 173.23 Muy1 : 110.31

INTERACTION RATIO: 0.11 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 21
END JOINT: 169 Puz : 3913.49 Muz : 284.41 Muy : 181.16 IR: 0.35

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            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    265  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:   164  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    179.60   Muy1 :    114.30

INTERACTION RATIO: 0.20 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    25
END JOINT:   170 Puz :   3913.49   Muz :    275.39   Muy :    175.27   IR: 0.31
================================================================================


================================================================================
            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    266  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    3 END JOINT:   165  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.50 Muy1 : 114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 171 Puz : 3913.49 Muz : 277.32 Muy : 176.50 IR: 0.31
==========

==========
IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 267 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 166 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.46 Muy1 : 114.83

INTERACTION RATIO: 0.20 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 172 Puz : 3913.49 Muz : 277.32 Muy : 176.50 IR: 0.31

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================================================================================


================================================================================
            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    268  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    3 END JOINT:   167  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.50   Muy1 :    114.85

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    25
END JOINT:   173 Puz :   3913.49   Muz :    275.39   Muy :    175.27   IR: 0.31
================================================================================


================================================================================
            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    269  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)               Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    2 END JOINT:   174  SHORT COLUMN
```

REQD. STEEL AREA : 355.10 Sq.mm.
REQD. CONCRETE AREA: 44387.97 Sq.mm.
MAIN REINFORCEMENT : Provide 8 - 12 dia. (0.38%, 904.78 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3345.73 Muz1 : 45.36 Muy1 : 30.38

INTERACTION RATIO: 0.96 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 25
END JOINT: 174 Puz : 3509.40 Muz : 149.52 Muy : 96.35 IR: 0.58
==========

==========
IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 270 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 169 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 178.12 Muy1 : 113.37

INTERACTION RATIO: 0.03 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 21
END JOINT: 175 Puz : 3913.49 Muz : 233.46 Muy : 147.53 IR: 0.39

```
================================================================================


================================================================================
            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    271  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    1 END JOINT:   170  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.11   Muy1 :    114.61

INTERACTION RATIO: 0.10 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    22
END JOINT:   176 Puz :   3913.49   Muz :    243.76   Muy :    154.33   IR: 0.29
================================================================================


================================================================================
            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    272  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    3 END JOINT:   171  TENSION COLUMN
```

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.50 Muy1 : 114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 22
END JOINT: 177 Puz : 3913.49 Muz : 244.60 Muy : 154.88 IR: 0.27
==========

==========
IS-456 L I M I T S T A T E D E S I G N
C O L U M N N O. 273 D E S I G N R E S U L T S

M30 Fe415 (Main) Fe415 (Sec.)

LENGTH: 3000.0 mm CROSS SECTION: 400.0 mm X 600.0 mm COVER: 40.0 mm

** GUIDING LOAD CASE: 1 END JOINT: 172 TENSION COLUMN

REQD. STEEL AREA : 1920.00 Sq.mm.
REQD. CONCRETE AREA: 238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide 20 - 12 dia. (0.94%, 2261.95 Sq.mm.)
(Equally distributed)
TIE REINFORCEMENT : Provide 8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)

Puz : 3811.68 Muz1 : 180.46 Muy1 : 114.83

INTERACTION RATIO: 0.11 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)

WORST LOAD CASE: 20
END JOINT: 178 Puz : 3913.49 Muz : 244.60 Muy : 154.88 IR: 0.27

```
================================================================================


================================================================================
            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    274  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:    3 END JOINT:   173  TENSION COLUMN


REQD. STEEL AREA   :     1920.00 Sq.mm.
REQD. CONCRETE AREA:   238080.00 Sq.mm.
MAIN REINFORCEMENT : Provide  20 - 12 dia. (0.94%,   2261.95 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3811.68   Muz1 :    180.50   Muy1 :    114.86

INTERACTION RATIO: 0.00 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    20
END JOINT:   179 Puz :   3913.49   Muz :    243.76   Muy :    154.33   IR: 0.29
================================================================================


================================================================================
            IS-456    L I M I T    S T A T E    D E S I G N
         C O L U M N   N O.    275  D E S I G N   R E S U L T S

      M30                    Fe415 (Main)              Fe415 (Sec.)

LENGTH:  3000.0 mm   CROSS SECTION:  400.0 mm X  600.0 mm  COVER: 40.0 mm

** GUIDING LOAD CASE:   21 END JOINT:   180  SHORT COLUMN
```

```
REQD. STEEL AREA   :      629.37 Sq.mm.
REQD. CONCRETE AREA:    78670.66 Sq.mm.
MAIN REINFORCEMENT : Provide   8 - 12 dia. (0.38%,    904.78 Sq.mm.)
                     (Equally distributed)
TIE REINFORCEMENT  : Provide  8 mm dia. rectangular ties @ 190 mm c/c

SECTION CAPACITY BASED ON REINFORCEMENT REQUIRED (KNS-MET)
----------------------------------------------------------
Puz :   3427.39   Muz1 :     90.44   Muy1 :     59.26

INTERACTION RATIO: 0.99 (as per Cl. 39.6, IS456:2000)

SECTION CAPACITY BASED ON REINFORCEMENT PROVIDED (KNS-MET)
----------------------------------------------------------
WORST LOAD CASE:    21
END JOINT:   180 Puz :   3509.40   Muz :    115.37   Muy :     74.71   IR: 0.78
==========================================================================


********************END OF COLUMN DESIGN RESULTS********************


255. DESIGN ELEMENT 396 TO 495
```

ELEMENT DESIGN SUMMARY

ELEMENT	LONG. REINF (SQ.MM/ME)	MOM-X /LOAD (KN-M/M)	TRANS. REINF (SQ.MM/ME)	MOM-Y /LOAD (KN-M/M)
396 TOP :	156.	0.49 / 20	156.	0.79 / 21
BOTT:	156.	-0.23 / 26	156.	-0.65 / 27
397 TOP :	156.	0.16 / 21	156.	0.80 / 21
BOTT:	156.	-0.11 / 27	156.	-0.68 / 27
398 TOP :	156.	0.14 / 21	156.	0.80 / 21
BOTT:	156.	-0.11 / 27	156.	-0.68 / 27
399 TOP :	156.	0.16 / 21	156.	0.80 / 21
BOTT:	156.	-0.11 / 27	156.	-0.68 / 27
400 TOP :	156.	0.49 / 22	156.	0.79 / 21
BOTT:	156.	-0.23 / 24	156.	-0.65 / 27
401 TOP :	156.	0.51 / 20	156.	0.10 / 20
BOTT:	156.	-0.26 / 26	156.	-0.04 / 26
402 TOP :	156.	0.04 / 23	156.	0.08 / 23
BOTT:	156.	-0.01 / 2	156.	-0.05 / 25
403 TOP :	156.	0.02 / 23	156.	0.08 / 23
BOTT:	156.	-0.01 / 2	156.	-0.05 / 25
404 TOP :	156.	0.04 / 23	156.	0.08 / 23
BOTT:	156.	-0.01 / 2	156.	-0.05 / 25
405 TOP :	156.	0.51 / 22	156.	0.10 / 22
BOTT:	156.	-0.26 / 24	156.	-0.04 / 1
406 TOP :	156.	0.51 / 20	156.	0.10 / 20
BOTT:	156.	-0.26 / 26	156.	-0.04 / 26
407 TOP :	156.	0.04 / 21	156.	0.08 / 21
BOTT:	156.	-0.01 / 1	156.	-0.05 / 27
408 TOP :	156.	0.02 / 21	156.	0.08 / 21
BOTT:	156.	-0.01 / 27	156.	-0.05 / 27
409 TOP :	156.	0.04 / 21	156.	0.08 / 21
BOTT:	156.	-0.00 / 4	156.	-0.05 / 27
410 TOP :	156.	0.51 / 22	156.	0.10 / 22
BOTT:	156.	-0.26 / 24	156.	-0.04 / 1
411 TOP :	156.	0.49 / 20	156.	0.79 / 23
BOTT:	156.	-0.23 / 26	156.	-0.65 / 25

412 TOP :	156.	0.16 / 23	156.	0.80 / 23
BOTT:	156.	-0.11 / 25	156.	-0.68 / 25
413 TOP :	156.	0.14 / 23	156.	0.80 / 23
BOTT:	156.	-0.11 / 25	156.	-0.68 / 25
414 TOP :	156.	0.16 / 23	156.	0.80 / 23
BOTT:	156.	-0.11 / 25	156.	-0.68 / 25
415 TOP :	156.	0.49 / 22	156.	0.79 / 23
BOTT:	156.	-0.23 / 24	156.	-0.65 / 25
416 TOP :	156.	0.40 / 20	156.	0.64 / 21
BOTT:	156.	-0.21 / 26	156.	-0.53 / 27
417 TOP :	156.	0.16 / 21	156.	0.67 / 21
BOTT:	156.	-0.07 / 27	156.	-0.57 / 27
418 TOP :	156.	0.12 / 21	156.	0.66 / 21
BOTT:	156.	-0.09 / 27	156.	-0.57 / 27
419 TOP :	156.	0.16 / 21	156.	0.67 / 21
BOTT:	156.	-0.07 / 27	156.	-0.57 / 27
420 TOP :	156.	0.40 / 22	156.	0.64 / 21
BOTT:	156.	-0.21 / 24	156.	-0.53 / 27
421 TOP :	156.	0.43 / 20	156.	0.10 / 20
BOTT:	156.	-0.23 / 26	156.	-0.02 / 26
422 TOP :	156.	0.07 / 20	156.	0.06 / 7
BOTT:	156.	-0.00 / 2	156.	-0.00 / 2
423 TOP :	156.	0.02 / 7	156.	0.05 / 7
BOTT:	156.	-0.00 / 2	156.	-0.00 / 2
424 TOP :	156.	0.07 / 22	156.	0.06 / 7
BOTT:	156.	-0.01 / 1	156.	-0.00 / 1
425 TOP :	156.	0.43 / 22	156.	0.10 / 22
BOTT:	156.	-0.23 / 24	156.	-0.03 / 1
426 TOP :	156.	0.43 / 20	156.	0.10 / 20
BOTT:	156.	-0.23 / 26	156.	-0.02 / 26
427 TOP :	156.	0.07 / 20	156.	0.06 / 7
BOTT:	156.	0.00 / 0	156.	-0.00 / 3
428 TOP :	156.	0.02 / 7	156.	0.05 / 7
BOTT:	156.	-0.00 / 3	156.	-0.00 / 3
429 TOP :	156.	0.07 / 22	156.	0.06 / 7
BOTT:	156.	-0.01 / 1	156.	-0.00 / 1

430	TOP :	156.	0.43 / 22	156.	0.10 / 22
	BOTT:	156.	-0.23 / 24	156.	-0.03 / 1
431	TOP :	156.	0.40 / 20	156.	0.64 / 23
	BOTT:	156.	-0.21 / 26	156.	-0.53 / 25
432	TOP :	156.	0.16 / 23	156.	0.67 / 23
	BOTT:	156.	-0.07 / 25	156.	-0.57 / 25
433	TOP :	156.	0.12 / 23	156.	0.66 / 23
	BOTT:	156.	-0.09 / 25	156.	-0.57 / 25
434	TOP :	156.	0.16 / 23	156.	0.67 / 23
	BOTT:	156.	-0.07 / 25	156.	-0.57 / 25
435	TOP :	156.	0.40 / 22	156.	0.64 / 23
	BOTT:	156.	-0.21 / 24	156.	-0.53 / 25
436	TOP :	156.	0.40 / 20	156.	0.61 / 21
	BOTT:	156.	-0.18 / 26	156.	-0.48 / 27
437	TOP :	156.	0.16 / 21	156.	0.62 / 21
	BOTT:	156.	-0.06 / 27	156.	-0.52 / 27
438	TOP :	156.	0.11 / 21	156.	0.62 / 21
	BOTT:	156.	-0.08 / 27	156.	-0.52 / 27
439	TOP :	156.	0.16 / 21	156.	0.62 / 21
	BOTT:	156.	-0.06 / 27	156.	-0.52 / 27
440	TOP :	156.	0.40 / 22	156.	0.61 / 21
	BOTT:	156.	-0.18 / 24	156.	-0.48 / 27
441	TOP :	156.	0.42 / 20	156.	0.11 / 20
	BOTT:	156.	-0.20 / 26	156.	-0.01 / 26
442	TOP :	156.	0.09 / 20	156.	0.08 / 7
	BOTT:	156.	-0.00 / 4	156.	0.00 / 0
443	TOP :	156.	0.02 / 7	156.	0.07 / 7
	BOTT:	156.	-0.00 / 4	156.	-0.00 / 1
444	TOP :	156.	0.09 / 22	156.	0.08 / 7
	BOTT:	156.	-0.01 / 1	156.	-0.00 / 1
445	TOP :	156.	0.42 / 22	156.	0.11 / 22
	BOTT:	156.	-0.20 / 24	156.	-0.03 / 1
446	TOP :	156.	0.42 / 20	156.	0.11 / 20
	BOTT:	156.	-0.20 / 26	156.	-0.01 / 26
447	TOP :	156.	0.09 / 20	156.	0.08 / 7
	BOTT:	156.	-0.00 / 2	156.	-0.01 / 2

448	TOP :	156.	0.02 / 7	156.	0.07 / 7
	BOTT:	156.	-0.00 / 2	156.	-0.01 / 2
449	TOP :	156.	0.09 / 22	156.	0.08 / 7
	BOTT:	156.	-0.01 / 1	156.	-0.01 / 2
450	TOP :	156.	0.42 / 22	156.	0.11 / 22
	BOTT:	156.	-0.20 / 24	156.	-0.03 / 1
451	TOP :	156.	0.40 / 20	156.	0.61 / 23
	BOTT:	156.	-0.18 / 26	156.	-0.48 / 25
452	TOP :	156.	0.16 / 23	156.	0.62 / 23
	BOTT:	156.	-0.06 / 2	156.	-0.52 / 25
453	TOP :	156.	0.11 / 23	156.	0.62 / 23
	BOTT:	156.	-0.08 / 25	156.	-0.52 / 25
454	TOP :	156.	0.16 / 23	156.	0.62 / 23
	BOTT:	156.	-0.06 / 2	156.	-0.52 / 25
455	TOP :	156.	0.40 / 22	156.	0.61 / 23
	BOTT:	156.	-0.18 / 24	156.	-0.48 / 25
456	TOP :	156.	0.27 / 20	156.	0.43 / 21
	BOTT:	156.	-0.11 / 26	156.	-0.33 / 27
457	TOP :	156.	0.14 / 21	156.	0.45 / 21
	BOTT:	156.	-0.02 / 27	156.	-0.35 / 27
458	TOP :	156.	0.09 / 21	156.	0.45 / 21
	BOTT:	156.	-0.05 / 27	156.	-0.35 / 27
459	TOP :	156.	0.14 / 21	156.	0.45 / 21
	BOTT:	156.	-0.02 / 27	156.	-0.35 / 27
460	TOP :	156.	0.27 / 22	156.	0.43 / 21
	BOTT:	156.	-0.11 / 1	156.	-0.33 / 27
461	TOP :	156.	0.29 / 20	156.	0.09 / 21
	BOTT:	156.	-0.13 / 26	156.	-0.00 / 4
462	TOP :	156.	0.11 / 20	156.	0.10 / 21
	BOTT:	156.	-0.00 / 3	156.	-0.00 / 4
463	TOP :	156.	0.03 / 21	156.	0.10 / 21
	BOTT:	156.	-0.00 / 3	156.	-0.00 / 27
464	TOP :	156.	0.11 / 22	156.	0.10 / 21
	BOTT:	156.	-0.02 / 1	156.	-0.00 / 1
465	TOP :	156.	0.29 / 22	156.	0.09 / 21
	BOTT:	156.	-0.13 / 24	156.	-0.02 / 1

466	TOP :	156.	0.29 / 20	156.	0.09 / 23
	BOTT:	156.	-0.13 / 26	156.	-0.02 / 2
467	TOP :	156.	0.11 / 20	156.	0.10 / 23
	BOTT:	156.	-0.00 / 2	156.	-0.03 / 2
468	TOP :	156.	0.03 / 23	156.	0.10 / 23
	BOTT:	156.	-0.00 / 2	156.	-0.03 / 2
469	TOP :	156.	0.11 / 22	156.	0.10 / 23
	BOTT:	156.	-0.02 / 1	156.	-0.03 / 2
470	TOP :	156.	0.29 / 22	156.	0.09 / 23
	BOTT:	156.	-0.13 / 24	156.	-0.02 / 2
471	TOP :	156.	0.27 / 20	156.	0.43 / 23
	BOTT:	156.	-0.11 / 26	156.	-0.33 / 25
472	TOP :	156.	0.14 / 23	156.	0.45 / 23
	BOTT:	156.	-0.04 / 2	156.	-0.35 / 25
473	TOP :	156.	0.09 / 23	156.	0.45 / 23
	BOTT:	156.	-0.05 / 25	156.	-0.35 / 25
474	TOP :	156.	0.14 / 23	156.	0.45 / 23
	BOTT:	156.	-0.04 / 2	156.	-0.35 / 25
475	TOP :	156.	0.27 / 22	156.	0.43 / 23
	BOTT:	156.	-0.11 / 1	156.	-0.33 / 25
476	TOP :	156.	0.47 / 20	156.	0.42 / 21
	BOTT:	156.	-0.02 / 26	156.	-0.23 / 27
477	TOP :	156.	0.17 / 21	156.	0.40 / 21
	BOTT:	156.	-0.00 / 3	156.	-0.26 / 27
478	TOP :	156.	0.08 / 21	156.	0.39 / 21
	BOTT:	156.	-0.04 / 27	156.	-0.26 / 27
479	TOP :	156.	0.17 / 21	156.	0.40 / 21
	BOTT:	156.	-0.03 / 1	156.	-0.26 / 27
480	TOP :	156.	0.47 / 22	156.	0.42 / 21
	BOTT:	156.	-0.12 / 1	156.	-0.23 / 27
481	TOP :	156.	0.47 / 20	156.	0.19 / 21
	BOTT:	156.	-0.03 / 26	156.	-0.01 / 27
482	TOP :	156.	0.19 / 20	156.	0.19 / 21
	BOTT:	156.	-0.00 / 3	156.	-0.02 / 27
483	TOP :	156.	0.04 / 21	156.	0.17 / 21
	BOTT:	156.	-0.00 / 3	156.	-0.03 / 27

484 TOP :	156.	0.19 /	22	156.	0.19 /	21
BOTT:	156.	-0.03 /	1	156.	-0.02 /	27
485 TOP :	156.	0.47 /	22	156.	0.19 /	21
BOTT:	156.	-0.13 /	1	156.	-0.02 /	1
486 TOP :	156.	0.47 /	20	156.	0.19 /	23
BOTT:	156.	-0.03 /	26	156.	-0.05 /	2
487 TOP :	156.	0.19 /	20	156.	0.19 /	23
BOTT:	156.	-0.01 /	2	156.	-0.06 /	2
488 TOP :	156.	0.04 /	23	156.	0.17 /	23
BOTT:	156.	-0.01 /	2	156.	-0.06 /	2
489 TOP :	156.	0.19 /	22	156.	0.19 /	23
BOTT:	156.	-0.03 /	1	156.	-0.06 /	2
490 TOP :	156.	0.47 /	22	156.	0.19 /	23
BOTT:	156.	-0.13 /	1	156.	-0.05 /	2
491 TOP :	156.	0.47 /	20	156.	0.42 /	23
BOTT:	156.	-0.03 /	2	156.	-0.23 /	25
492 TOP :	156.	0.17 /	23	156.	0.40 /	23
BOTT:	156.	-0.03 /	2	156.	-0.26 /	25
493 TOP :	156.	0.08 /	23	156.	0.39 /	23
BOTT:	156.	-0.04 /	25	156.	-0.26 /	25
494 TOP :	156.	0.17 /	23	156.	0.40 /	23
BOTT:	156.	-0.03 /	2	156.	-0.26 /	25
495 TOP :	156.	0.47 /	22	156.	0.42 /	23
BOTT:	156.	-0.12 /	1	156.	-0.23 /	25

**********END OF ELEMENT DESIGN**********

256. CONCRETE TAKE
257. END CONCRETE DESIGN

********** CONCRETE TAKE OFF **********
(FOR BEAMS, COLUMNS AND PLATES DESIGNED ABOVE)

NOTE: CONCRETE QUANTITY REPRESENTS VOLUME OF CONCRETE IN BEAMS, COLUMNS, AND PLATES DESIGNED ABOVE.
REINFORCING STEEL QUANTITY REPRESENTS REINFORCING STEEL IN BEAMS AND COLUMNS DESIGNED ABOVE.
REINFORCING STEEL IN PLATES IS NOT INCLUDED IN THE REPORTED QUANTITY.

TOTAL VOLUME OF CONCRETE = 227.8 CU.METER

BAR DIA (in mm)	WEIGHT (in New)
8	37945
10	31708
12	85391
16	2647
*** TOTAL=	157692

258. PERFORM ANALYSIS

```
*******************************************************
* UNITS - KN    METE                                  *
* TIME PERIOD FOR X 1893 LOADING =    0.58020 SEC     *
* SA/G PER 1893=    2.344, LOAD FACTOR= 1.000         *
* VB PER 1893=    0.0375 X    27847.70=    1044.40 KN  *
*                                                     *
*******************************************************
```

```
*******************************************************
* UNITS - KN    METE                                  *
* TIME PERIOD FOR Z 1893 LOADING =    0.58020 SEC     *
* SA/G PER 1893=    2.344, LOAD FACTOR= 1.000         *
* VB PER 1893=    0.0375 X    27847.70=    1044.40 KN  *
*                                                     *
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259. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= AUG 1,2026 TIME= 23:31:52 ****

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* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
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